

On localization phenomena in plasticity models

Bernardo Ardini [2146900]

Padova, 5 febbraio 2026

Contents

I	Necking of a metallic cylindrical bar	4
1	Small deformation model	5
1.1	Formulation	5
1.1.1	Motivation	5
1.1.2	Kinematics	6
1.1.3	Statics	6
1.1.4	Constitutive relations	7
1.2	Existence and uniqueness result	8
1.2.1	Function spaces and hypotheses	8
1.2.2	Formulation of the problem	8
1.2.3	Existence and uniqueness theorem	8
1.3	Some analytical solutions	10
1.3.1	Traction with free end sections	10
1.3.2	Traction with clamped end sections	10
II	Shear band localization in cohesive materials	15
2	Small deformation model	16
2.1	Formulation	16
2.1.1	Kinematics	16
2.1.2	Statics	16
2.1.3	Dissipation inequality	18
2.1.4	Constitutive relations	18
2.2	Mathematical analysis	19
2.2.1	Hypotheses	19
2.2.2	Function spaces	20
2.2.3	Formulation of the problem	20
2.2.4	Global existence	20
2.2.5	Example of constitutive relations	31
2.2.6	Further regularity hypotheses	32
2.2.7	Local existence and uniqueness and global uniqueness	32
2.2.8	Bifurcation analysis	34
2.3	Numerical approximation	38
3	Large deformation model	39
3.1	Formulation	39
3.1.1	Kinematics	39
3.1.2	Statics	39

A	Notation	41
A.1	Tensors	41
B	Modeling in continuum mechanics	42
C	Tools from functional analysis	43
C.1	Linear functional analysis	43
C.1.1	Other	43
C.1.2	The Aubin–Lions–Simon lemma	43
C.1.3	Generalized mixed elliptic problems	43
C.2	Nonlinear functional analysis	47
C.2.1	Other	47
C.2.2	Implicit function theorem	48
C.2.3	Fixed point theorems	48
C.2.4	Local existence theorem for Cauchy problems in Banach spaces . . .	48
C.2.5	Browder–Minty theorem	49

Summary of what I already done and plan for what I would like to do

Introduction

The thesis should be concerned with the study of a strain localization in a plasticity model. To make the analysis more tractable it would be better if the localization is captured also by a small strain model. In the field of metal plasticity the necking is an example of strain localization. Since Von Mises plasticity with hardening at small strain does not exhibit bifurcation (uniqueness and stability are guaranteed) then to detect the necking instability it is necessary to consider the geometric nonlinearity. Instead in materials whose yield strength depends on the pressure (for example granular materials), localization is evident also in the small strain regime (the formation of shear bands). This behavior is for example exhibited by the nonassociative Drucker–Prager model [2], also at small strain and in presence of hardening. For that reason I decided to focus my attention to that model.

What I have already done

Instead of adopting the usual formulation of the nonassociative Drucker–Prager model I reformulated it in an alternative way for the following two reasons

- *Mechanical reason.* The Drucker–Prager model is formulated by formally adopting the same framework used in metal plasticity to materials whose yield strength depend on pressure. The difference from metal plasticity is that volumetric plastic strains are allowed and the yield region is chosen in the space Sym of all symmetric stress tensors instead of DevSym . This means that the deviatoric and volumetric plastic response are described as a unique plasticity mechanism. This is actually nonphysical since while the deviatoric plastic response is a friction mechanism (that increases with the pressure, a sort of internal Coulomb friction), the change in the volumetric plastic strain (the dilatancy) is due to a kinematical constrain (the grains must distance themselves from each other to be able to slide over each other). Since the Drucker–Prager model associative overestimates dilatancy the nonassociative behavior is introduced. This ad hoc modification is a clear sign that the deviatoric plastic response should be treated independently from the changes in the volumetric plastic strain instead.
- *Mathematical reason.* Non associative plasticity is difficult to be handled mathematically since it has a not evident variational structure [15].

In the alternative model the volumetric plastic strain is kinematically constrained (for simplicity I have considered only the case in which it constrained to zero¹, i.e., the material

¹Actually I started to study also the general case but I still finish.

is non dilatant), a Lagrange multiplier (a sort of “plastic pressure” that in practice coincides with the macroscopic pressure) is introduced to impose that constrain, and the yield strength is made constitutively dependent on the microscopic pressure. To formulate the model I started from the framework of plastic strain gradient viscoplasticity of Gurtin and Anand (developed for metals) based on the principle of virtual powers and on macroforce and microforce balances.

I have proved the existence of a global in time solution using the Rothe method (up to errors in the proof). First we need to solve the discrete in time problem. At each time step there is a nonlinear mixed elliptic (saddle point) problem to solve (it is mixed due to the presence of the constrain). It is solved by analogy with the linear mixed elliptic problem: first the constrained is solved, then knowing the constrained space we solve the leading elliptic problem, then the Lagrange multiplier is determined. The leading elliptic problem is solved using the Minty–Browder theorem (the minimization works as well but it fails when the material is dilatant because the potential structure is lost, for that reason I opted for the Minty–Browder theorem). Due to the pressure dependent yield strength the leading elliptic problem gives a pressure dependent solution. The existence of the pressure is obtained with the Schauder fixed point theorem. The need of compactness, necessary to apply the Schauder fixed point theorem, obliged us to add a further regularization term depending on the gradient of the macroscopic strain (in this way we have constructed a second gradient mechanical model [21]). Maybe without the second gradient regularization the solution could be found in $BD(\Omega)$ has happen in [12] but I decided to not investigate this possibility. To get the time continuous solution we need to pass to the vanishing time step limit. This is done by first obtaining a priori estimates on the kinematical variables and their rates and then weakly convergent subsequences are extracted. To handle the nonlinear dependence on the pressure in the limit passage, we establish the strong convergence of the pressure using the Aubin–Lions lemma. To get the a priori estimates on the rates I added both a macroscopic and a microscopic viscosity.

Assuming further regularity on the constitutive relations I obtained local existence and uniqueness of solutions (again up to error in the proof). This is done locally recasting the problem as an ODE using the implicit function theorem and then proving the local existence, uniqueness and regularity in time using the Cauchy–Lipschitz–Picard theorem. This ensure that the global solution is unique and regular in time.

What I would like to do

Since viscosity makes the solution unique the system does not have bifurcation. However we expect that for small viscosity the solution became strongly dependent on the problem data. In particular small perturbations of the data leads to a completely different localization pattern. For example if the shape of a cylindrical specimen in compression is not perfectly symmetric the shear bands that appear are not symmetric at all. In the vanishing viscosity limit the problems admits multiple solution and jumps in the time evolution.

I would like to understand better the phenomenon of localization trying to obtain some qualitative or quantitative insight. Until now I have studied the stiffness operator in absence of viscosity (the Fréchet derivative of the residual functional that arise at each step in the Rothe discretization) and I have clues to claim that it tends to become singular at a certain stage of the time evolution if hardening is small enough, suggesting that the evolution experience a bifurcation or a jump. In Section 2.2.8 I have reported more considerations and details. In particular the computation reported there reveal how the dependence of the yield strength on the plastic pressure seems to make our problem more prone to

instabilities then Von Mises plasticity where that dependence is not present. Anyway I have no other ideas how to make clearer that analysis. If it is useful I will try to obtain the existence of solutions in absence of viscosity. Since the solution probably has jumps we cannot prove the existence without some selection criteria. A possibility is to adopt the notion of energetic solution, but I would prefer to avoid it because it is nonphysical, being based on a global minimization at each time step. Moreover the saddle point structure of the problem complicates the situation. What I would prefer to do is to study the limit with vanishing viscosity, perhaps trying to understand and apply to this model the time rescaling procedure used in [12] and [24]. However, if it is possible, I would like to keep a positive viscosity and

- study the sensitivity of the solution to perturbation of the data (in some sense to be understood),
- make clear the fact that during localization there are some modes that evolves faster (the shear band that is quickly forming) and other modes that evolves slower (the distributed deformation), maybe one understand this looking at the spectrum of the stiffness operator and look the eigenspace associated to the smaller eigenvalue, or the kernel (maybe it is finite dimensional) of the stiffness operator without viscosity when it became singular.

I would like also to confirm the above considerations by a numerical study of the problem using the finite element method. The proof of the existence theorem is close to the numerical scheme that would be used, so we have in part established the convergence of the numerical method. I would use the monolithic approach proposed in [28] and to obviate the need of globally C^1 finite element to handle the second gradient I would adopt the mortar method (we use discontinuous Lagrange polynomial of order two and we impose weakly the continuity and the continuity of the gradient across the elements with Lagrange multipliers supported on the element edges representing tractions and torques exchanged between the elements)². If the viscosity is kept positive the problem could be solved with a standard Newton–Raphson procedure applying a time step size adaptivity procedure to slow down during the fast localization process. This should work until the viscosity is so small that the problem requires a too small time step to be solved.

²I saw this method applied for the numerical treatment of Navier–Stokes- $\alpha\beta$ model of Gurtin and Fried but I cannot find the paper any more.

Part I

Necking of a metallic cylindrical bar

Chapter 1

Small deformation model

1.1 Formulation

1.1.1 Motivation

Let $\Omega = (0, L) \times \Sigma$ with $L > 0$ and $\Sigma \subset \mathbb{R}^{d-1}$ be the reference configuration of the cylindrical bar. We denote the generic point in Ω as (x, y) with $x \in (0, L)$ and $y \in \Sigma$. We assume a macroscopic displacement of the form

$$\begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} u_1(x) \\ u_2(x)y \end{pmatrix}$$

where u_1 and u_2 are functions from $(0, L)$ to $\mathbb{R}$. This assumption means that the sections of the cylindrical bar remain flat and can translate and expand uniformly. The corresponding displacement gradient is

$$H = \begin{pmatrix} u_1' & 0 \\ u_2' y & u_2 I \end{pmatrix}.$$

The plastic distortion is assumed to be of the form

$$H_p = \begin{pmatrix} p & 0 \\ 0 & -\frac{p}{d-1} I \end{pmatrix}$$

where $p: (0, L) \rightarrow \mathbb{R}$. This choice implies plastic incompressibility. The elastic distortion is then given by

$$H_e = H - H_p = \begin{pmatrix} u_1' - p & 0 \\ u_2' y & (u_2 + \frac{p}{d-1}) I \end{pmatrix}.$$

Then the elastic strain is

$$E_e = \begin{pmatrix} u_1' - p & \frac{u_2'}{2} y^T \\ \frac{u_2'}{2} y & (u_2 + \frac{p}{d-1}) I \end{pmatrix} = \begin{pmatrix} e_1 & \frac{u_2'}{2} y^T \\ \frac{u_2'}{2} y & e_2 I \end{pmatrix}$$

where we have introduced $e_1 = u_1' - p$ and $e_2 = u_2 + (d-1)^{-1}p$. Assuming an isotropic elastic response the elastic free energy density is given by

$$\psi_e = \frac{1}{2} \kappa (\text{tr } E_e)^2 + \mu |\text{dev } E_e|^2$$

where $\kappa \geq 0$ and $\mu \geq 0$ are the bulk and shear modulus respectively. We notice that

$$\text{tr } E_e = e_1 + (d-1)e_2 \quad \text{and} \quad \text{dev } E_e = \begin{pmatrix} \frac{d-1}{d}(e_1 - e_2) & \frac{u_2'}{2} y^T \\ \frac{u_2'}{2} y & -\frac{1}{d}(e_1 - e_2)I \end{pmatrix}.$$

Then the elastic free energy density can be written as

$$\psi_e = \frac{1}{2}\kappa(e_1 + (d-1)e_2)^2 + \frac{d-1}{d}\mu(e_1 - e_2)^2 + \frac{1}{2}|y|^2\mu|u'_2|^2.$$

We conclude that the elastic free energy per unit axial length is

$$\int_{\Sigma} \psi_e dy = \frac{1}{2}A\kappa(e_1 + (d-1)e_2)^2 + \frac{d-1}{d}A\mu(e_1 - e_2)^2 + \frac{1}{2}J\mu|u'_2|^2$$

where

$$A = \int_{\Sigma} dy \quad \text{and} \quad J = \int_{\Sigma} |y|^2 dy$$

are the polar and the polar second order moment of area.

1.1.2 Kinematics

Let $L > 0$ be the length of the cylindrical bar. The macroscopic configuration of the bar is described by $u_i : (0, L) \rightarrow \mathbb{R}$ for $i \in \{1, 2\}$. Here u_1 is the displacement of the sections in the axis direction and u_2 is the quantity describing the expansion of the sections. The configuration of the microscopic structure of the bar is described by $p : (0, L) \rightarrow \mathbb{R}$. We define $e_1 = u'_1 - p$ and $e_2 = u_2 + (d-1)^{-1}p$.

1.1.3 Statics

We denote with $\tilde{v}$ and $\tilde{q}$ the virtual rates of u and p respectively. To describe the coupling between the macroscopic configuration u and the microscopic configuration p we choose an internal virtual power of the form

$$\tilde{W} = \int_0^L \zeta \tilde{v}'_2 dx + \int_0^L (\sigma_1(\tilde{v}'_1 - \tilde{q}) + \sigma_2(\tilde{v}_2 + (d-1)^{-1}\tilde{q})) dx + \int_0^L \tau \tilde{q} dx.$$

The first integral describes a purely macroscopic response and it is introduced to gain regularity of u_2 . Here ζ is the transversal stress. The second integral describes the elastic response that couples the macroscopic and the microscopic configurations. Here σ_1 is the longitudinal elastic stress and σ_2 the longitudinal elastic force. The last integral describes the microscopic response and τ is the microscopic force. The external longitudinal and transversal forces distributed along the length of the bar and the external longitudinal and transversal forces acting on the extremities of the bar are included choosing the external power

$$\tilde{\mathcal{P}} = \int_0^L f \cdot \tilde{v} dx + h(L) \cdot \tilde{v}(L) - h(0) \cdot \tilde{v}(0).$$

Definition 1.1 (Principle of Virtual Powers). Equilibrium holds when $\tilde{W} + \tilde{\mathcal{P}} = \tilde{\mathcal{R}}$ for all virtual rates.

Lemma 1.2. The local form of the equilibrium condition is equivalent to

$$\sigma'_1 + f_1 = 0 \quad \text{in } (0, L) \tag{1.1.1a}$$

$$h_1(x) = \sigma_1(x) \quad \text{for } x \in \{0, L\} \tag{1.1.1b}$$

$$-\sigma_2 + \zeta' + f_2 = 0 \quad \text{in } (0, L) \tag{1.1.1c}$$

$$h_2(x) = \zeta(x) \quad \text{for } x \in \{0, L\} \tag{1.1.1d}$$

$$\sigma_1 - (d-1)^{-1}\sigma_2 - \tau = 0 \quad \text{in } (0, L) \tag{1.1.1e}$$

Proof. First we choose $\tilde{v}_2 = 0$ and $\tilde{q} = 0$ and $\tilde{v}_1$ arbitrary. Then we get

$$-\int_0^L \sigma_1' \tilde{v}_1 dx + \sigma_1(L)\tilde{v}_1(L) - \sigma_1(0)\tilde{v}_1(0) = \int_0^L f_1 \tilde{v}_1 dx + h_1(L)\tilde{v}_1(L) - h_1(0)\tilde{v}_1(0).$$

We get (1.1.1a) and (1.1.1b). Next we choose $\tilde{v}_1 = 0$ and $\tilde{q} = 0$ and $\tilde{v}_2$ arbitrary. Then

$$-\int_0^L \zeta' \tilde{v}_2 dx + \zeta(L)\tilde{v}_2(L) - \zeta(0)\tilde{v}_2(0) + \int_0^L \sigma_2 \tilde{v}_2 dx = \int_0^L f_2 \tilde{v}_2 dx + h_2(L)\tilde{v}_2(L) - h_2(0)\tilde{v}_2(0).$$

We get (1.1.1c) and (1.1.1d). Then we choose $\tilde{v} = 0$ and $\tilde{q}$ arbitrary obtaining

$$\int_0^L (\sigma_1 - (d-1)^{-1}\sigma_2 - \tau)\tilde{q} dx = 0.$$

This implies (1.1.1e). □

1.1.4 Constitutive relations

We choose the free energy for unit axial length

$$\psi = \frac{1}{2}\kappa(e_1 + (d-1)e_2)^2 + \frac{1}{2}\mu(e_1 - e_2)^2 + \frac{1}{2}l^2\mu|u_2'|^2 + \frac{1}{2}Bp^2.$$

where κ is a bulk modulus, μ is a shear modulus, l is length parameter related to the geometry of the section and B is the hardening parameter. We also choose the dissipation potential for unit axial length

$$\phi = R(\dot{p})$$

where $R: \mathbb{R} \rightarrow [0, \infty]$ satisfy $R(0) = 0$. A possible choice could be $R(\dot{p}) = Y|\dot{p}|$ where $Y > 0$ is the yield strength. Consistently with these choices we impose the following constitutive relations

$$\begin{aligned} \zeta &= \mu l^2 u_2' \\ \sigma_1 &= \kappa(e_1 + (d-1)e_2) + \mu(e_1 - e_2) \\ \sigma_2 &= (d-1)\kappa(e_1 + (d-1)e_2) - \mu(e_1 - e_2) \\ \tau &\in Bp + \partial R(\dot{p}). \end{aligned}$$

Lemma 1.3. Every process that obeys the constitutive relations satisfy the dissipation inequality.

Proof. Let $\tau = Bp + \xi$ with $\xi \in \partial R(\dot{p})$. The dissipation per unit axial length is given by

$$\zeta \dot{u}_2' + \sigma_1 \dot{e}_1 + \sigma_2 \dot{e}_2 + \tau \dot{p} - \dot{\psi} = \xi \dot{p}.$$

By definition of subdifferential we have $0 = R(0) \geq R(\dot{p}) - \xi \dot{p}$ so that $\xi \dot{p} \geq R(\dot{p}) \geq 0$. This implies that the dissipation is nonnegative. □

We notice that introducing the constitutive relations into (1.1.1e) we get

$$\frac{d}{d-1}\mu(e_1 - e_2) - Bp \in \partial R(\dot{p}).$$

We refer to

$$m = \frac{d}{d-1}\mu(e_1 - e_2)$$

as the Mandel stress. It is the elastic force acting on the microscopic configuration and we notice that it depends on the shear modulus only.

1.2 Existence and uniqueness result

In this section we discuss the existence and uniqueness of solutions. For simplicity we will assume homogeneous Dirichlet boundary conditions $u(x) = 0$ for $x \in \{0, L\}$.

1.2.1 Function spaces and hypotheses

We define the spaces $\mathbf{U} = H^1((0, L), \mathbb{R} \times \mathbb{R})$ and $\mathbf{P} = L^2((0, L))$. Let also $\mathbf{Z} = \mathbf{U} \times \mathbf{P}$. We will write $z = (u, p) \in \mathbf{Z}$ with $u \in \mathbf{U}$ and $p \in \mathbf{P}$ and $w = (v, q) \in \mathbf{Z}$ with $v \in \mathbf{U}$ and $q \in \mathbf{P}$.

The existence and uniqueness result will be obtained under the following hypotheses.

- (G) $L > 0$ and $d = 2$ or $d = 3$
- (ψ) κ, μ, l and B are positive constants
- (R) $R: \mathbb{R} \rightarrow \mathbb{R}$ is given by $R(q) = Y|q|$ for some $Y > 0$
- (f) $f \in H^1((0, T), \mathbf{U}^*)$ and $f(0) = 0$

1.2.2 Formulation of the problem

We introduce the linear operator $\mathcal{A}: \mathbf{Z} \rightarrow \mathbf{Z}^*$ defined by

$$\begin{aligned} \langle \mathcal{A}z, w \rangle = & \int_0^L l^2 \mu u'_2 v'_2 dx + \int_0^L \left[\kappa(u'_1 + (d-1)u_2) + \mu \left(u'_1 - u_2 - \frac{d}{d-1} p \right) \right] (v'_1 - q) dx + \\ & + \int_0^L \left[(d-1)\kappa(u'_1 + (d-1)u_2) - \mu \left(u'_1 - u_2 - \frac{d}{d-1} q \right) \right] (v_2 + (d-1)^{-1} p) dx + \\ & + \int_0^L B p q dx \end{aligned}$$

and the functional $\mathcal{R}: \mathbf{Z} \rightarrow \mathbb{R} \cup \{\infty\}$ defined by

$$\mathcal{R}(w) = \int_0^L R(q) dx.$$

We define also $\mathcal{F}: [0, T] \rightarrow \mathbf{Z}^*$ by

$$\langle \mathcal{F}(t), w \rangle = \int_0^T f \cdot v dx.$$

Now the boundary value problem associated to the model introduced in Section 1.1 can be formulated as follows.

Problem 1.4. Find $z = (u, p) \in AC([0, T], \mathbf{Z})$ such that $z(0) = 0$ and for almost every $t \in (0, T)$

$$\langle \mathcal{A}z(t), w - \dot{z}(t) \rangle + \mathcal{R}(w) - \mathcal{R}(\dot{z}(t)) \geq \langle \mathcal{F}(t), w - \dot{z}(t) \rangle \quad \text{for all } w \in \mathbf{Z}.$$

1.2.3 Existence and uniqueness theorem

To prove the existence and uniqueness result we make use of the following general theorem.

Theorem 1.5. Let $\mathbf{Z}$ be a reflexive and separable Banach space. Let $\mathcal{A}: \mathbf{Z} \rightarrow \mathbf{Z}^*$ be a linear and continuous functional which is also coercive, meaning that

$$\frac{\langle \mathcal{A}z, z \rangle}{\|z\|_{\mathbf{Z}}} \rightarrow \infty \quad \text{as } \|z\|_{\mathbf{Z}} \rightarrow \infty.$$

Let $\mathcal{R}: \mathbf{Z} \rightarrow \mathbb{R} \cup \{\infty\}$ be convex, proper, l.s.c., 1-homogeneous and Lipchitz continuous on its domain. Let $\mathcal{F} \in H^1((0, T), \mathbf{Z}^*)$ such that $\mathcal{F}(0) = 0$. Then there exists a unique $z \in AC([0, T], \mathbf{Z})$ such that for almost every $t \in (0, T)$

$$\langle \mathcal{A}z(t), w - \dot{z}(t) \rangle + \mathcal{R}(w) - \mathcal{R}(\dot{z}(t)) \geq \langle \mathcal{F}(t), w - \dot{z}(t) \rangle \quad \text{for all } w \in \mathbf{Z}.$$

Proof. See [22]. □

The main result of this section is the following theorem.

Theorem 1.6. Assume the hypotheses of Section 1.2.1. Then there exists a unique solution to Problem 1.4.

Proof. We verify the hypothesis of Theorem 1.5.

Step 1. It is trivial to check that $\mathcal{A}$ is linear and continuous. To prove coercivity we show that there exists $\alpha > 0$ such that $\langle \mathcal{A}z, z \rangle \geq \alpha \|z\|_{\mathbf{Z}}^2$. If we call $e_1 = u'_1 - p$ and $e_2 = u_2 + (d-1)^{-1}p$ we have by applying Young inequality repeatedly

$$\begin{aligned} \langle \mathcal{A}z, z \rangle &= 2 \int_0^L \psi dx = \kappa \|e_1 + (d-1)e_2\|_2^2 + \mu \|e_1 - e_2\|_2^2 + l^2 \mu \|u'_2\|_2^2 + B \|p\|_2^2 = \\ &= \kappa \|u'_1 + (d-1)u_2\|_2^2 + \mu \left\| u'_1 - u_2 - \frac{d}{d-1} p \right\|_2^2 + l^2 \mu \|u'_2\|_2^2 + B \|p\|_2^2 \geq \\ &\geq (1-\theta) \kappa \|u'_1\|_2^2 + \left(1 - \frac{1}{\theta}\right) (d-1)^2 \kappa \|u_2\|_2^2 + (1-\tau) \mu \|u'_1 - u_2\|_2^2 + \\ &\quad l^2 \mu \|u'_2\|_2^2 + \left[\left(1 - \frac{1}{\tau}\right) \left(\frac{d}{d-1}\right)^2 \mu + B \right] \|p\|_2^2 \\ &\geq [(1-\theta) \kappa + (1-\eta)(1-\tau) \mu] \|u'_1\|_2^2 + \\ &\quad + \left[\left(1 - \frac{1}{\theta}\right) (d-1)^2 \kappa + \left(1 - \frac{1}{\eta}\right) (1-\tau) \mu \right] \|u_2\|_2^2 + l^2 \mu \|u'_2\|_2^2 + \\ &\quad + \left[\left(1 - \frac{1}{\tau}\right) \left(\frac{d}{d-1}\right)^2 \mu + B \right] \|p\|_2^2 \end{aligned}$$

for all θ, τ and η in $(0, \infty)$. We restrict to θ, τ and η in $(0, 1)$. Then by Poincarè inequality there exist $C_1 > 0$ and $C_2 > 0$ such that

$$\begin{aligned} \langle \mathcal{A}z, z \rangle &\geq C_1 [(1-\theta) \kappa + (1-\eta)(1-\tau) \mu] \|u_1\|_{1,2}^2 + \\ &\quad + \left[C_2 l^2 \mu + \left(1 - \frac{1}{\theta}\right) (d-1)^2 \kappa + \left(1 - \frac{1}{\eta}\right) (1-\tau) \mu \right] \|u_2\|_{1,2}^2 + \\ &\quad + \left[\left(1 - \frac{1}{\tau}\right) \left(\frac{d}{d-1}\right)^2 \mu + B \right] \|p\|_2^2 \end{aligned}$$

Now we choose θ, τ and η sufficiently close to 1 so that the coefficients multiplying $\|u_1\|_{1,2}^2$, $\|u_2\|_{1,2}^2$ and $\|p\|_2^2$ are positive and we get the desired coercivity.

Step 2. Since $\mathcal{R}(w) = Y \|q\|_1$ we see immediately that $\mathcal{R}$ is convex, proper, continuous and 1-homogeneous.

Step 3. Since $f \in H^1((0, T), \mathbf{U}^*)$ and $f(0) = 0$ then $\mathcal{F} \in H^1((0, T), \mathbf{Z}^*)$ and $\mathcal{F}(0) = 0$. □

1.3 Some analytical solutions

1.3.1 Traction with free end sections

Suppose that $f(x, t) = 0$, $h_1(L, t) = h_1(0, t) = F(t)$, and $h_2(L, t) = h_2(0, t) = 0$ for all $x \in (0, L)$ and $t \in [0, T]$. We assume that $F(0) = 0$ and that F is monotonically increasing. From equation (1.1.1a) and (1.1.1b) we have that $\sigma_1 = F$ on all $(0, L)$. Then

$$\kappa(e_1 + (d-1)e_2) = F - \mu(e_1 - e_2) \quad \text{and} \quad e_1 = \frac{F - (d-1)\kappa e_2 + \mu e_2}{\kappa + \mu}.$$

Moreover

$$e_1 - e_2 = \frac{F - d\kappa e_2}{\kappa + \mu}.$$

Then

$$\begin{aligned} \sigma_2 &= (d-1)(F - \mu(e_1 - e_2)) - \mu(e_1 - e_2) = (d-1)F - d\mu(e_1 - e_2) = \\ &= (d-1)F - d\frac{\mu}{\kappa + \mu}F + d^2\frac{\kappa\mu}{\kappa + \mu}e_2 = \frac{(d-1)\kappa - \mu}{\kappa + \mu}F + d^2\frac{\kappa\mu}{\kappa + \mu}e_2 = \\ &= \frac{(d-1)\kappa - \mu}{\kappa + \mu}F + \frac{d^2}{d-1}\frac{\kappa\mu}{\kappa + \mu}p + d^2\frac{\kappa\mu}{\kappa + \mu}u_2. \end{aligned}$$

Now from equation (1.1.1c) we have

$$l^2\mu u_2'' = \frac{(d-1)\kappa - \mu}{\kappa + \mu}F + \frac{d^2}{d-1}\frac{\kappa\mu}{\kappa + \mu}p + d^2\frac{\kappa\mu}{\kappa + \mu}u_2.$$

The solution is given by

$$u_2 = -\frac{p}{d-1} - \frac{(d-1)\kappa - \mu}{d^2\kappa\mu}F.$$

since conditions $h_2(L, t) = h_2(0, t)$ are satisfied because $u_2' = 0$. Notice that

$$e_2 = u_2 + \frac{p}{d-1} = -\frac{(d-1)\kappa - \mu}{d^2\kappa\mu}F.$$

Now the Mandel stress is given by

$$m = \frac{d}{d-1}\frac{\mu}{\kappa + \mu}(F - d\kappa e_2) = F.$$

When $F \leq Y$ then the response is purely elastic and $p = 0$. When $F > Y$ then $\dot{p} > 0$ and

$$F - Bp = Y.$$

This implies $p = B^{-1}(F - Y)$. Assuming $(d-1)\kappa - \mu > 0$ the sections constricts in traction.

1.3.2 Traction with clamped end sections

Suppose that $f(x, t) = 0$, $h_1(L, t) = h_1(0, t) = F(t)$, and $u_2(L, t) = u_2(0, t) = 0$ for all $x \in (0, L)$ and $t \in [0, T]$. We assume that $F(0) = 0$ and that F is monotonically increasing. By the same computations of Section 1.3.1 we find the equation

$$l^2\mu u_2'' = \frac{(d-1)\kappa - \mu}{\kappa + \mu}F + \frac{d^2}{d-1}\frac{\kappa\mu}{\kappa + \mu}p + d^2\frac{\kappa\mu}{\kappa + \mu}u_2. \quad (1.3.1)$$

In the elastic range $p = 0$ everywhere in $(0, L)$. In this case a particular solution is given by

$$x \mapsto -\frac{(d-1)\kappa - \mu}{d^2\kappa\mu} F$$

and a solution of the homogeneous equation which is also symmetric with respect to $x = L/2$ is

$$x \mapsto \cosh\left(d\sqrt{\frac{\kappa}{\kappa+\mu}} \frac{x - \frac{L}{2}}{l}\right).$$

Then taking into account the boundary conditions we find the solution

$$u_2 = e_2 = \frac{(d-1)\kappa - \mu}{d^2\kappa\mu} \left(\frac{\cosh\left(d\sqrt{\frac{\kappa}{\kappa+\mu}} \frac{x - \frac{L}{2}}{l}\right)}{\cosh\left(d\sqrt{\frac{\kappa}{\kappa+\mu}} \frac{L}{2l}\right)} - 1 \right) F.$$

When $(d-1)\kappa - \mu > 0$ the sections constricts in traction. The middle section is the most affected by the constriction. The Mandel stress in the elastic range is given by

$$m = \frac{d}{d-1} \frac{\mu}{\kappa + \mu} (F - d\kappa e_2) = \frac{d}{d-1} \frac{\mu}{\kappa + \mu} \left(1 - \frac{(d-1)\kappa - \mu}{d\mu} \left(\frac{\cosh\left(d\sqrt{\frac{\kappa}{\kappa+\mu}} \frac{x - \frac{L}{2}}{l}\right)}{\cosh\left(d\sqrt{\frac{\kappa}{\kappa+\mu}} \frac{L}{2l}\right)} - 1 \right) \right) F.$$

Its maximum is reached in the middle section so we conclude that the Mandel stress reach yield strength here first. Then we claim that the region where there is a nonzero plastic distortion is an interval $(L/2 - \epsilon, L/2 + \epsilon)$ where $\epsilon \geq 0$ is a function of time. Since the Mandel stress is positive we claim also that $p > 0$ in that interval. We now focus on the half of the bar corresponding to interval $(0, L/2)$. On $(0, L/2 - \epsilon)$ the solution is of the form

$$\begin{cases} u_2 = \frac{(d-1)\kappa - \mu}{d^2\kappa\mu} \left(\frac{\cosh\left(d\sqrt{\frac{\kappa}{\kappa+\mu}} \frac{x - \frac{L}{2}}{l}\right)}{\cosh\left(d\sqrt{\frac{\kappa}{\kappa+\mu}} \frac{L}{2l}\right)} - 1 \right) F + C_1 \sinh\left(d\sqrt{\frac{\kappa}{\kappa+\mu}} x\right) \\ p = 0 \end{cases}$$

where $C_1 \in \mathbb{R}$. On $(L/2 - \epsilon, L/2)$ we have

$$m - Bp = Y$$

$$\frac{d}{d-1} \frac{\mu}{\kappa + \mu} (F - d\kappa u_2 - \frac{d\kappa}{d-1} p) - Bp = Y.$$

Then we find

$$p = \frac{\frac{d}{d-1} \frac{\mu}{\kappa + \mu} F - \frac{d^2}{d-1} \frac{\kappa\mu}{\kappa + \mu} u_2 - Y}{\frac{d^2}{(d-1)^2} \frac{\mu\kappa}{\kappa + \mu} + B}. \quad (1.3.2)$$

Substituting that identity into equation (1.3.1) we have obtained a differential equation in y of the form

$$y'' = ay + b$$

where a and b are two constants. A solution which is symmetric with respect to $x = L/2$ is of the form

$$y = -\frac{b}{a} + C_2 \cosh\left(\sqrt{a} \left(x - \frac{L}{2}\right)\right).$$

Remark 1.7. Since the computations became too long to be done by hand we perform them with an automatic algebraic manipulator (the MATLAB Symbolic Math Toolbox). Since most of the resultant expressions are too long we do not report them.

We now show explicitly that there exists a solution $u_2 \in C^2([0, L])$. To this aim we impose the continuity of u_2 and u_2' at $x = L/2 - \epsilon$ and solving a linear system we find the constants C_1 and C_2 . We need only to find ϵ . To determine it we consider the Mandel stress computed the elastic range at $x = L/2$. We call F_Y the smallest value of F for which that Mandel stress reach Y . For $F \leq F_Y$ we have $\epsilon = \eta(F) = 0$. To determine ϵ when $F > F_Y$ we impose that $m(L/2 - \epsilon) = Y$. Since the resultant equation cannot be solved analytically in ϵ we find F as a smooth function $f =: [0, \epsilon_0) \rightarrow \mathbb{R}$ of ϵ . Here

$$\epsilon_0 = \sup_{F \in [0, \infty)} \eta(F).$$

We notice that by continuity $F_Y = f(0)$. We find

$$F_Y = \frac{d-1}{d} \frac{\kappa + \mu}{\mu} \left(-\frac{(d-1)\kappa - \mu}{d\mu} \left(\frac{1}{\cosh\left(d\sqrt{\frac{\kappa}{\kappa+\mu}} \frac{L}{2l}\right)} - 1 \right) + 1 \right)^{-1} Y.$$

After a meticulous analysis of the analytical expression of f we find that $\epsilon_0 > L/2$. This means that when the load reach the value $F = f(L/2)$ to the plastic distortion satisfies $p > 0$ in the entire interval $(0, L)$. An explicit expression of ϵ_0 is not available. To get further insight on the solution we look at the limit $l \rightarrow 0^+$.

Proposition 1.8. In the limit $l \rightarrow 0^+$ the solution u_2 that we have found converges at least pointwise on $(0, L) \times [0, T]$ to the homogeneous solution of Section 1.3.1. This convergence is also uniform on compact subsets of $(0, L)$.

Proof. It is a direct computation. □

Further information on the solution can be obtained looking at particular values of the parameters. We fix $d = 3$, $L = 1$, $l = 0.1$, $\kappa = 100$, $\mu = 100$, $B = 0.1$, $Y = 1$. In Figure 1.1 on the left we reported the graph of f . We see that ϵ_0 , which correspond to the abscissa of the vertical asymptote, is slightly greater then $L/2$. On the right we report as a function of ϵ the value of u_2 computed at $x = L/2$ and $x = L/100$ that means at the middle and near the end of the bar respectively. This last plot suggest that the shrinkage of the cross section has no limit and happens everywhere along the bar. In particular we have evidences for the following claim.

Claim 1.9. The solution u_2 converges to $-\infty$ uniformly on compact sets contained in $(0, L)$.

Remark 1.10. From the information obtained so far, we conclude that the small deformation model is not able to predict necking. Indeed the shrinkage of the cross section seems to be not localized on a small region around the middle of the bar but to be distributed along its entire length.

For completeness in Figure 1.2 we report the plot of u_2 and p when $\epsilon = 0.3$. In Figure 1.3 we report a picture of the bar in the same situation. On the top there is the result obtained with a purely elastic bar while on the top the result predicted including the plastic behavior. The displacements of cross sections is magnified with a factor 7. The color represents the variable p .

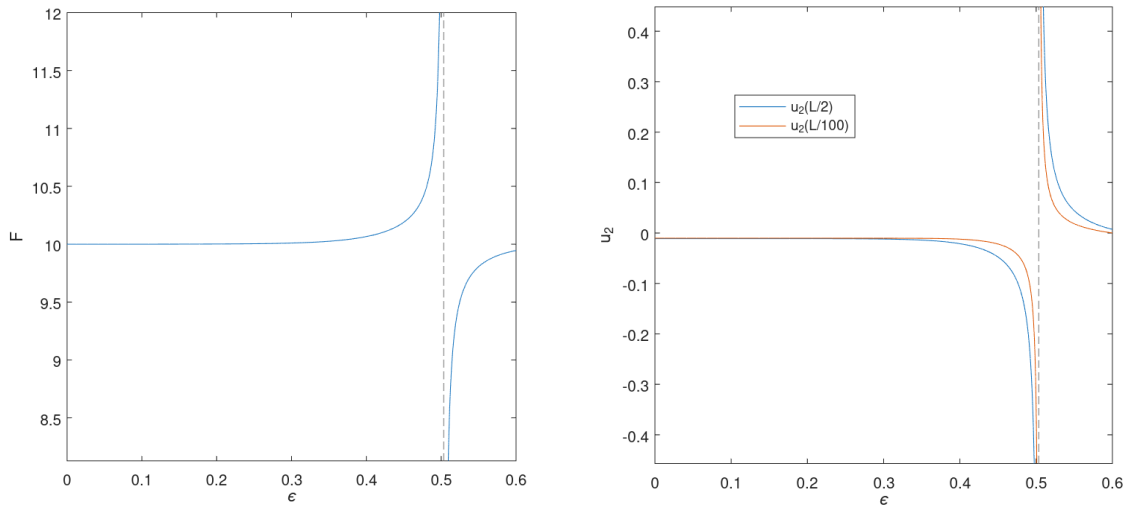


Figure 1.1: The plot on the left show the function f . The plot on the right show the value of u_2 at $x = L/2$ (at the middle of the bar) and $x = L/100$ (near the end of the bar).

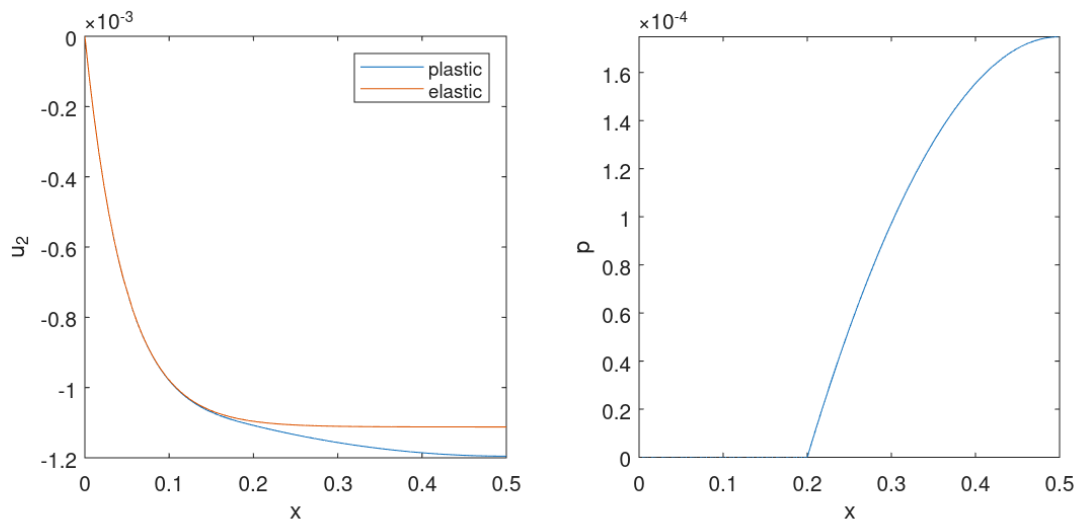


Figure 1.2: Plot of u_2 (left) and p (right) for $\epsilon = 0.3$.

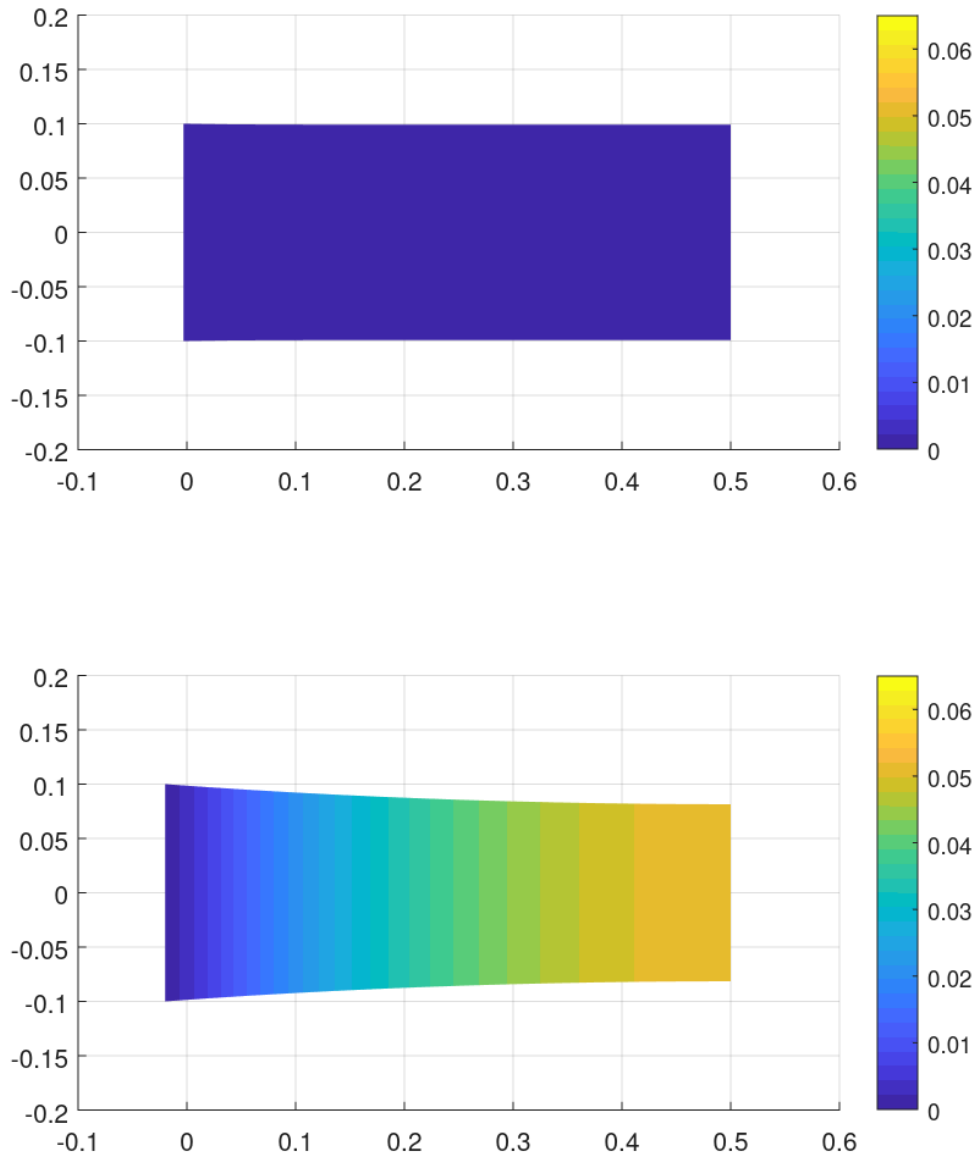


Figure 1.3: Picture of the half $(0, L/2)$ of the bar when $\epsilon = 0.493$. The bar on the bottom is plastic, the one on the top is purely elastic and subject to the same load. The transversal displacement is magnified with a factor 7. The color represents p .

Part II

Shear band localization in cohesive materials

Chapter 2

Small deformation model

2.1 Formulation

In this section we formulate the mechanical model. The description is made very quickly and we will expand it to give a better justification. In this section the emphasis is on the mechanical modeling, the correct functional setting will be investigated in the subsequent sections.

2.1.1 Kinematics

Let $\Omega \subset \mathbb{R}^d$ be the reference configuration of the body. The macroscopic displacement is a map $u: \Omega \rightarrow \mathbb{R}^d$. We set $E = \text{sym grad } u$ and $M = \text{grad}^2 u$. We subdivide $\partial\Omega$ in two disjoint parts Γ_D and Γ_N . On Γ_D we impose the kinematical constrain $u = \bar{u}$. The configuration of the microscopic structure is described by the plastic distortion $H_p: \Omega \rightarrow \text{Lin}$. The skew symmetric part $W_p = \text{skew } H_p$ describes the rotation of the microscopic structure and for simplicity we assume that it vanish. The symmetric part is the plastic strain $E_p = \text{sym } H_p: \Omega \rightarrow \text{Sym}$ which describes the deformation of microscopic structure. We can regard E_p as the elastically relaxed strain of the body so we define the elastic strain as $E_e = E - E_p$. The volumetric deformation of the microscopic structure is described by $\lambda_p = \text{tr } E_p$ while the deviatoric deformation by $U_p = \text{dev } E_p$. We call $M_p = \text{grad } U_p$ and we set $\mathbb{U}_p = (U_p, l M_p) \in \text{DevPla}$ where $l > 0$ is an internal length scale. Here $\text{DevPla} = \text{Sym} \otimes (\text{Sym} \otimes \mathbb{R})$ is the vector space of the deviatoric plastic strains and their gradients. In what follows we will denote with $\circ$ the natural scalar product on DevPla . We assume that the material is not dilatant (this is a simplification and should be generalized to describe the dilatant behavior observed in most materials exhibiting a pressure dependent yield strength) so we add the constrain $\lambda_p = 0$. To impose this constrain we add the Lagrange multiplier $p: \Omega \rightarrow \mathbb{R}$ with the meaning of a plastic pressure. The constrain can be written in weak form as

$$\int_{\Omega} \lambda_p \tilde{q} dx = 0 \quad \text{for all } \tilde{q}: \Omega \rightarrow \mathbb{R}.$$

2.1.2 Statics

We denote with $\tilde{v}$, $\tilde{L}_p$, $\tilde{V}_p$ and $\tilde{\mu}_p$ the virtual rates of u , E_p , U_p and λ_p respectively. The virtual rate of E_e is $\tilde{L}_e = \text{grad } \tilde{v} - \tilde{L}_p$ where $\tilde{L} = \text{grad } \tilde{v}$. Due to the presence of the Dirichlet condition we require that $\tilde{v}$ vanish on Γ_D . Instead we do not require that $\tilde{\mu}_p = 0$ because the constrain $\lambda_p = 0$ is imposed by mean of the Lagrange multiplier p . To describe the coupling

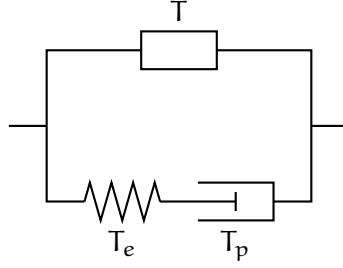


Figure 2.1: Rheological model describing the choice of the internal virtual power $\tilde{W}$. The box in upper branch represents the (generic) macroscopic response. In the lower branch, the spring represents the elastic response and the piston represents the microscopic response. We notice that the microscopic response besides the usual dissipative plastic microforce force can include conservative microforces to model hardening. A more detailed discussion on constitutive relations is in Section 2.1.4.

between the macroscopic deformation u and the microscopic configuration E_p we choose an internal virtual power of the form

$$\tilde{W} = \int_{\Omega} \left(T : \tilde{L} + K : \text{grad}^2 \tilde{v} \right) dx + \int_{\Omega} T_e : \tilde{L}_e dx + \int_{\Omega} \left(T_p : \tilde{V}_p + K_p : \text{grad} \tilde{V}_p \right) dx.$$

The first integral is related to the macroscopic response. Here $T \in \text{Sym}$ is the macrostress and $K \in \text{Lin} \otimes \mathbb{R}$ is the macrohyperstress. The presence of the second order of $\tilde{v}$ is due to the need of a sufficient regularization in view of the existence theorem. The second integral describe the coupling between the macroscopic and the microscopic response and $T_e \in \text{Sym}$ is the elastic stress. The third integral is associated with the microscopic response. Here $T_p \in \text{DevSym}$ is the microforce and $K_p \in \text{DevSym} \otimes \mathbb{R}$ is the microstress. In Figure 2.1 we report a picture that represents schematically the meaning of the internal virtual power that we have chosen. We define $\tilde{V}_p = (\tilde{V}_p, l \text{grad} \tilde{V}_p)$ and $T_p = (T_p, l^{-1} K_p)$ so that

$$T_p : \tilde{V}_p + K_p : \text{grad} \tilde{V}_p = T_p \circ \tilde{V}_p.$$

The action of the plastic pressure is obtained by considering the reaction virtual power

$$\tilde{\mathcal{R}} = - \int_{\Omega} p \tilde{\mu}_p dx.$$

We notice that this power vanish on virtual rates that satisfy the constrain that is $\tilde{\mu}_p = 0$. The external actions of the volume force f on Ω and the traction h on Γ_N are included choosing the external virtual power

$$\tilde{\mathcal{P}} = \int_{\Omega} f \cdot \tilde{v} dx + \int_{\Gamma_N} h \cdot \tilde{v} da.$$

For simplicity we assume that no external torque is applied to the body and that there is no external action on the microscopic configuration. (Maybe these external actions could be added to describe better the boundary conditions imposed on the body.)

Definition 2.1 (Principle of Virtual Powers). Equilibrium holds when $\tilde{W} + \tilde{\mathcal{P}} = \tilde{\mathcal{R}}$ for all virtual rates.

To get the local form of the equilibrium condition we apply the principle of virtual powers.

Remark. I reported the local form of the equilibrium in the case $K = 0$ only. I will report the general case following [21].

Lemma 2.2. The local form of the equilibrium condition is

$$\begin{aligned} \operatorname{div}(\mathbf{T} + \mathbf{T}_e) + \mathbf{f} &= 0 \quad \text{in } \Omega \\ (\mathbf{T} + \mathbf{T}_e)\mathbf{n} &= \mathbf{h} \quad \text{in } \Gamma_N \\ -\operatorname{dev} \mathbf{T}_e + \mathbf{T}_p - \operatorname{div} \mathbf{K}_p &= 0 \quad \text{in } \Omega \\ p &= -\frac{1}{d} \operatorname{tr} \mathbf{T}_e \quad \text{in } \Omega \\ \mathbf{K}_p \mathbf{n} &= 0 \quad \text{in } \partial\Omega. \end{aligned}$$

Proof. We can write the virtual power as

$$\begin{aligned} \tilde{\mathcal{W}} = - \int_{\Omega} \operatorname{div}(\mathbf{T} + \mathbf{T}_e) : \tilde{\mathbf{v}} \, dx &+ \int_{\Gamma_N} (\mathbf{T} + \mathbf{T}_e)\mathbf{n} \cdot \tilde{\mathbf{v}} \, dx + \int_{\Omega} (\mathbf{T}_p - \operatorname{dev} \mathbf{T}_e - \operatorname{div} \mathbf{K}_p) : \tilde{\mathbf{V}}_p \, dx + \\ &+ \int_{\partial\Omega} \mathbf{K}_p \mathbf{n} : \tilde{\mathbf{V}}_p \, dx. \end{aligned}$$

Then the results follows from the fundamental theorem of the calculus of variations. $\square$

2.1.3 Dissipation inequality

Let ψ be the free energy per unit volume in the reference configuration. Then the free energy contained in a portion Λ of Ω is

$$\Psi = \int_{\Lambda} \psi \, dx.$$

Let now $\mathcal{W}$ the internal power absorbed by the portion Λ . We define the dissipation rate in Λ as

$$\mathcal{D} = \mathcal{W} - \dot{\Psi}.$$

The dissipation inequality impose that $\mathcal{D} \geq 0$ for all Λ .

2.1.4 Constitutive relations

We assume a constitutive relation for the free energy of the form

$$\psi = \frac{1}{2} \mathfrak{C} \mathbf{E}_e : \mathbf{E}_e + \frac{1}{2} \mathfrak{A} \mathbf{M} : \mathbf{M} + \frac{1}{2} \mathfrak{B} \mathbf{U}_p \circ \mathbf{U}_p.$$

with $\mathfrak{C}$, $\mathfrak{A}$ and $\mathfrak{B}$ a symmetric and positive definite linear operators on Sym , $\operatorname{Lin} \otimes \mathbb{R}$ and DevPla respectively. The macroscopic response is described by

$$\mathbf{T} = \mathfrak{D} \dot{\mathbf{E}} \quad \text{and} \quad \mathbf{K} = \mathfrak{A} \dot{\mathbf{M}} + \mathfrak{E} \dot{\mathbf{M}}.$$

with $\mathfrak{D}$ and $\mathfrak{E}$ symmetric and positive definite linear operators on Sym and $\operatorname{Lin} \otimes \mathbb{R}$ respectively. The elastic response is given by

$$\mathbf{T}_e = \mathfrak{C} \mathbf{E}_e$$

and the viscoplastic response by

$$\mathbf{T}_p = \mathfrak{B} \mathbf{U}_p + \mathfrak{F} \dot{\mathbf{U}}_p + Y(p) \partial_{\dot{\mathbf{U}}_p} G(\dot{\mathbf{U}}_p)$$

where $\mathfrak{F}$ is a symmetric positive definite tensor on DevPla , $G : \operatorname{DevSym} \rightarrow \mathbb{R}$ is the plastic dissipation potential satisfying $\partial_{\dot{\mathbf{U}}_p} G(\dot{\mathbf{U}}_p) : \dot{\mathbf{U}}_p \geq 0$ and $Y : \mathbb{R} \rightarrow [0, \infty)$ is the pressure dependent yield strength.

Remark 2.3. In the above expression we are viewing $Y(p)\partial_{\dot{U}_p} G(\dot{U}_p) \in \text{DevSym}$ as an element of DevPla by considering DevSym as a subspace of DevPla .

Lemma 2.4. The constitutive relations satisfy the dissipation inequality.

Proof. The rate of the free energy per unit volume is

$$\dot{\psi} = \mathfrak{C}E_e : \dot{E}_e + \mathfrak{A}M : \text{grad}^2 \dot{u} + \mathfrak{B}U_p \circ \dot{U}_p$$

We notice that

$$T_p : \dot{U}_p + K_p : \text{grad} \dot{U}_p = \mathbb{T}_p \circ \dot{U}_p$$

Then the internal power per unit volume is

$$\begin{aligned} \mathfrak{D}\dot{E} : \dot{E} + \mathfrak{A}M : \text{grad}^2 \dot{u} + \mathfrak{E} \text{grad}^2 \dot{u} : \text{grad}^2 \dot{u} + \mathfrak{C}E_e : \dot{E}_e + \\ + \mathfrak{B}U_p \circ \dot{U}_p + \mathfrak{F}\dot{U}_p \circ \dot{U}_p + Y(p)\partial_{\dot{U}_p} G(\dot{U}_p) : \dot{U}_p. \end{aligned}$$

Taking the difference we find that the dissipation rate per unit volume is

$$\mathfrak{D}\dot{E} : \dot{E} + \mathfrak{F}\dot{U}_p \circ \dot{U}_p + Y(p)\partial_{\dot{U}_p} G(\dot{U}_p) : \dot{U}_p \geq 0.$$

This concludes the proof. $\square$

2.2 Mathematical analysis

In what follows we write simply U and λ in place of U_p and λ_p respectively. Moreover for simplicity we assume $\Gamma_D = \partial\Omega$ and $\bar{u} = 0$ since the extension to the general case is trivial.

2.2.1 Hypotheses

To get the global existence result we assume the following hypotheses.

- (S) (a) $\Omega \subset \mathbb{R}^d$ with $d \in \{2, 3\}$ is open, bounded and Lipschitz
(b) Γ_D and Γ_N are relatively open disjoint subsets of $\partial\Omega$
- (Y) (a) $Y \in C^{0,1}(\mathbb{R}, [0, \infty))$
- (G) (a) $G \in C^{1,1}(\text{DevSym}, \mathbb{R})$ is convex
(b) $\partial G \in L^\infty(\text{DevSym}, \text{DevPla})$
(c) $\partial G(\dot{U}) : \dot{U} \geq 0$ for all $\dot{U} \in \text{DevSym}$
- (L) (a) $f \in C^1([0, T], L^2(\Omega)^d)$
(b) $h \in C^1([0, T], L^2(\Gamma_N)^d)$
- (E) (a) $\mathfrak{C}$ and $\mathfrak{D}$ are in $\text{Sym} \otimes \text{Sym}$, $\mathfrak{B}$ and $\mathfrak{F}$ are in $\text{DevPla} \otimes \text{DevPla}$, $\mathfrak{A}$ and $\mathfrak{E}$ are in $(\text{Lin} \otimes \mathbb{R}) \otimes (\text{Lin} \otimes \mathbb{R})$
(b) there exist $\gamma > 0$ and $\delta > 0$ such that $\mathfrak{C}E : E \geq \gamma|E|^2$ and $\mathfrak{D}E : E \geq \delta|E|^2$ for all $E \in \text{Sym}$, there exist $\beta > 0$ and ϕ such that $\mathfrak{B}U \circ U \geq \beta|U|^2$ and $\mathfrak{F}U \circ U \geq \phi|U|^2$, there exists $\alpha > 0$ and $\epsilon > 0$ such that $\mathfrak{A}M : M \geq \alpha|\text{sym } M|^2$ and $\mathfrak{E}M : M \geq \epsilon|\text{sym } M|^2$ for all $M \in \text{Lin} \otimes \mathbb{R}$ where sym denote the symmetrization of the first two indices

2.2.2 Function spaces

We now introduce the function spaces that we use to formulate the problem. Let

$$\mathcal{Z} = \left\{ y = (u, U, \lambda) \left| \begin{array}{l} u \in H^2(\Omega)^d \text{ and } u = 0 \text{ on } \Gamma_D \\ U \in H^1(\Omega, \text{DevSym}) \\ \lambda \in L^2(\Omega) \end{array} \right. \right\} \quad \text{and} \quad \mathcal{Q} = \{ p \in L^2(\Omega) \}.$$

The rates of $y = (u, U, \lambda) \in \mathcal{Z}$ will be denoted by $\dot{y} = (\dot{u}, \dot{U}, \dot{\lambda}) \in \mathcal{Z}$. The variations of $\dot{y} = (\dot{u}, \dot{\xi}, \dot{\lambda}) \in \mathcal{Z}$ and $p \in \mathcal{Q}$ will be denoted as $z = (v, V, \mu) \in \mathcal{Z}$ and $q \in \mathcal{Q}$. The test functions will be denoted as $\tilde{z} = (\tilde{v}, \tilde{V}, \tilde{\mu})$.

2.2.3 Formulation of the problem

We define $\mathcal{F} : \mathcal{Z} \times \mathcal{Q} \times \mathcal{Z} \times [0, T] \rightarrow \mathcal{Z}' \times \mathcal{Q}'$ by

$$\begin{aligned} \langle \mathcal{F}(\dot{y}, p, y, t), (\tilde{z}, \tilde{q}) \rangle = & \int_{\Omega} \mathfrak{C} \left(\mathbb{E}u - U - \frac{\lambda}{d} I \right) : \left(\mathbb{E}\tilde{v} - \tilde{V} - \frac{\tilde{\mu}}{d} I \right) dx + \\ & + \int_{\Omega} (\mathfrak{A} \text{grad}^2 u + \mathfrak{E} \text{grad}^2 \dot{u}) : \text{grad}^2 \tilde{v} dx + \int_{\Omega} \mathfrak{D} \mathbb{E} \dot{u} : \mathbb{E} \tilde{v} dx + \\ & + \int_{\Omega} [(\mathfrak{B}U + \mathfrak{F}\dot{U}) \circ \tilde{V} + Y(p) \partial G(\dot{U}) : \tilde{V}] dx - \\ & - \int_{\Omega} p \tilde{\mu} dx - \int_{\Omega} \lambda \tilde{q} dx - \\ & - \int_{\Omega} f(t) \cdot \tilde{v} dx - \int_{\Gamma_N} h(t) \cdot \tilde{v} dx. \end{aligned}$$

Lemma 2.5. The map $\mathcal{F}$ is well defined.

Proof. The linear terms are trivially well defined so we need to check only

$$\int_{\Omega} Y(p) \partial G(\dot{U}) : \tilde{V} dx. \quad (2.2.1)$$

But

$$\int_{\Omega} |Y(p)| |\partial G(\dot{U})| |\tilde{V}| dx \leq \| \partial G \|_{\infty} \int_{\Omega} (Y(0) + \text{Lip}(Y)|p|) |\tilde{V}| dx \lesssim (1 + \|p\|_2) \|\tilde{V}\|_{1,2}.$$

This concludes the proof. $\square$

Now we can formulate the evolution problem as follows.

Problem 2.6. Find $y \in H^1((0, T), \mathcal{Z})$ and $p \in L^2([0, T], \mathcal{Q})$ such that $y(0) = 0$ and

$$\mathcal{F}(\dot{y}(t), p(t), y(t), t) = 0$$

for almost every $t \in (0, T)$.

2.2.4 Global existence

This section is devoted to prove the following existence of a global in time solution to Problem 2.6.

Theorem 2.7. Assume the hypotheses of Section 2.2.1. Then there exists $y \in H^1((0, T), \mathcal{Z})$ and $p \in C^0([0, T], \mathcal{Q})$ solution to Problem 2.6.

The proof is based on the Rothe method so we first prove the existence to a discrete in time version of the problem and then we pass to the vanishing time step limit.

Existence of the discrete in time solution

Let $N \in \mathbb{N}$. We define $\tau_N = N^{-1}T$ and $t_n^N = n\tau_N$ for $n \in \{0, \dots, N\}$. We consider the discrete in time version of Problem 2.6.

Problem 2.8 (Discrete problem). Let $y_0^N = 0 \in \mathcal{Z}$. Find $y_n^N \in \mathcal{Z}$ and $p_n^N \in \mathcal{Q}$ for $n \in \{1, \dots, N\}$ such that

$$\mathcal{F}\left(\frac{y_n^N - y_{n-1}^N}{\tau_N}, p_n^N, y_n^N, t_n^N\right) = 0 \quad (2.2.2)$$

for all $n \in \{1, \dots, N\}$.

We first prove the existence of a solution to Problem 2.8. Reasoning by induction we reduce to find $y_n^N \in \mathcal{Z}$ and $p_n^N \in \mathcal{Q}$ that satisfy (2.2.2) given $y_{n-1}^N \in \mathcal{Z}$. We immediately observe that in view of the constrain $\lambda_n^N = 0$ for all n . To simplify the notation we call $y = y_{n-1}^N$, $\dot{y} = \tau^{-1}(y_n^N - y_{n-1}^N)$ and $t = t_n^N$. Then $y_n^N = y + \tau\dot{y}$. It is enough to prove that there exists $\dot{y} \in \mathcal{Z}$ and $p \in \mathcal{Q}$ such that

$$\mathcal{F}(\dot{y}, p, y + h\dot{y}, t) = 0. \quad (2.2.3)$$

We define the operator $\mathcal{A}: \mathcal{Q} \times \mathcal{Z} \rightarrow \mathcal{Z}'$ by

$$\begin{aligned} \langle \mathcal{A}(p, \dot{y}), \tilde{z} \rangle &= \int_{\Omega} \tau \mathfrak{C} \left(E\dot{u} - \dot{U} - \frac{\dot{\lambda}}{d} I \right) : \left(E\tilde{v} - \tilde{V} - \frac{\tilde{\mu}}{d} I \right) dx + \int_{\Omega} \mathfrak{D} E\dot{u} : E\tilde{v} dx + \\ &+ \int_{\Omega} (\mathfrak{E} + \tau \mathfrak{A}) \text{grad}^2 \dot{u} : \text{grad}^2 \tilde{v} dx + \int_{\Omega} [(\mathfrak{F} + \tau \mathfrak{B}) \dot{U} \circ \tilde{V} + Y(p) \partial G(\dot{U}) : \tilde{V}] dx \end{aligned}$$

the linear operator $\mathcal{B}: \mathcal{Q} \rightarrow \mathcal{Z}'$ by

$$\langle \mathcal{B}p, \tilde{z} \rangle = - \int_{\Omega} p \tilde{\mu} dx$$

and the operator $\mathcal{C}: \mathcal{Z} \rightarrow \mathcal{Q}'$ by

$$\langle \mathcal{C}(\dot{y}), \tilde{q} \rangle = \int_{\Omega} (\lambda + \tau \dot{\lambda}) \tilde{q} dx = 0.$$

Moreover we introduce the linear form $l \in \mathcal{Z}'$

$$\begin{aligned} \langle l, \tilde{z} \rangle &= \int_{\Omega} f(t) \cdot \tilde{v} dx + \int_{\Gamma_N} h(t) \cdot \tilde{v} dx - \\ &- \int_{\Omega} \mathfrak{C} \left(E\dot{u} - \dot{U} - \frac{\dot{\lambda}}{d} I \right) : \left(E\tilde{v} - \tilde{V} - \frac{\tilde{\mu}}{d} I \right) dx - \\ &- \int_{\Omega} \mathfrak{A} \text{grad}^2 \dot{u} : \text{grad}^2 \tilde{v} dx - \int_{\Omega} \mathfrak{B} \dot{U} \circ \tilde{V} dx. \end{aligned}$$

Now equation (2.2.3) can be written as

$$\begin{cases} \mathcal{A}(p, \dot{y}) + \mathcal{B}p = l(t) \\ \mathcal{C}(\dot{y}) = 0. \end{cases}$$

We have already noticed that $\lambda = 0$ and $\dot{\lambda} = 0$. We call

$$\mathcal{N} = \left\{ \dot{y} = (\dot{u}, \dot{U}, 0) \mid \begin{array}{l} \dot{u} \in H^2(\Omega)^d \text{ and } \dot{u} = 0 \text{ on } \Gamma_D \\ \dot{U} \in H^1(\Omega, \text{DevSym}) \end{array} \right\}.$$

Now we study the first equation in (2.2.3). We first study the solvability of that equation in $\dot{\mathbf{y}} \in \mathcal{N}$ given $\mathbf{p} \in \mathcal{Q}$. If we test the first equation with $\tilde{\mathbf{z}} \in \mathcal{N}$ we get

$$\langle \mathcal{A}(\mathbf{p}, \dot{\mathbf{y}}), \tilde{\mathbf{z}} \rangle = \langle \mathbf{l}, \tilde{\mathbf{z}} \rangle \quad \text{for all } \tilde{\mathbf{z}} \in \mathcal{N}.$$

Now we define the space

$$\mathcal{X} = \left\{ \xi = (\dot{\mathbf{u}}, \dot{\mathbf{U}}) \mid \begin{array}{l} \dot{\mathbf{u}} \in H^2(\Omega)^d \text{ and } \dot{\mathbf{u}} = 0 \text{ on } \Gamma_D \\ \dot{\mathbf{U}} \in H^1(\Omega, \text{DevSym}) \end{array} \right\}$$

and we can consider the natural bijection $\mathcal{J}$ from $\mathcal{X}$ to $\mathcal{N}$. In particular if $\xi = (\dot{\mathbf{u}}, \dot{\mathbf{U}})$ we define

$$\mathcal{J}(\xi) = (\dot{\mathbf{u}}, \dot{\mathbf{U}}, 0).$$

Now we introduce the operator $\mathcal{T}_p$ from $\mathcal{X}$ to $\mathcal{X}'$ as follows. Given $\xi = (\dot{\mathbf{u}}, \dot{\mathbf{U}})$ and $\tilde{\xi} = (\tilde{\mathbf{v}}, \tilde{\mathbf{V}})$ in $\mathcal{X}$ we set

$$\begin{aligned} \langle \mathcal{T}_p(\xi), \tilde{\xi} \rangle &= \langle \mathcal{A}(\mathbf{p}, \mathcal{J}(\xi)), \mathcal{J}(\tilde{\xi}) \rangle = \int_{\Omega} \tau \mathfrak{E}(\mathbf{E}\dot{\mathbf{u}} - \dot{\mathbf{U}}) : (\mathbf{E}\tilde{\mathbf{v}} - \tilde{\mathbf{V}}) \, dx + \int_{\Omega} \mathfrak{D}\mathbf{E}\dot{\mathbf{u}} : \mathbf{E}\tilde{\mathbf{v}} \, dx + \\ &+ \int_{\Omega} (\mathfrak{E} + \tau \mathfrak{A}) \, \text{grad}^2 \dot{\mathbf{u}} : \text{grad}^2 \tilde{\mathbf{v}} \, dx + \int_{\Omega} [(\mathfrak{F} + \tau \mathfrak{B})\dot{\mathbf{U}} \circ \tilde{\mathbf{V}} + Y(\mathbf{p})\partial G(\dot{\mathbf{U}}) : \tilde{\mathbf{V}}] \, dx. \end{aligned}$$

Now for each $\mathbf{p} \in \mathcal{Q}$ we are reduced to find $\xi \in \mathcal{X}$ such that

$$\langle \mathcal{T}_p(\xi), \tilde{\xi} \rangle = \langle \mathbf{l}, \mathcal{J}(\tilde{\xi}) \rangle \quad \text{for all } \tilde{\xi} \in \mathcal{X}. \quad (2.2.4)$$

We want to apply the Browder–Minty theorem (Theorem C.16). We need to establish the boundedness, continuity, coercivity and monotonicity of $\mathcal{T}_p$. This done in the following lemmas.

Lemma 2.9. The operator $\mathcal{T}_p : \mathcal{X} \rightarrow \mathcal{X}'$ is bounded and continuous.

Proof. We have

$$\begin{aligned} |\langle \mathcal{T}_p(\xi), \tilde{\xi} \rangle| &\lesssim (\|\mathbf{E}\dot{\mathbf{u}}\|_2 + \|\dot{\mathbf{U}}\|_2) (\|\mathbf{E}\tilde{\mathbf{v}}\|_2 + \|\tilde{\mathbf{V}}\|_2) + \\ &+ \|\text{grad}^2 \dot{\mathbf{u}}\|_2 \|\text{grad}^2 \tilde{\mathbf{v}}\|_2 + \|\dot{\mathbf{U}}\|_2 \|\tilde{\mathbf{V}}\|_2 + (1 + \|\mathbf{p}\|_2) \|\tilde{\mathbf{V}}\|_2. \end{aligned}$$

and this proves that $\mathcal{T}_p$ is bounded. Now it is enough to prove the continuity of the nonlinear term only. Let $\xi_n \rightarrow \xi$ in $\mathcal{X}$. Then $\dot{\mathbf{U}}_n \rightarrow \dot{\mathbf{U}}$ in L^2 . We have

$$\left| \int_{\Omega} Y(\mathbf{p})(\partial G(\dot{\mathbf{U}}_n) - \partial G(\dot{\mathbf{U}})) \circ \tilde{\mathbf{V}} \, dx \right| \leq \|Y(\mathbf{p})(\partial G(\dot{\mathbf{U}}_n) - \partial G(\dot{\mathbf{U}}))\|_2 \|\tilde{\mathbf{V}}\|_2$$

and

$$\|Y(\mathbf{p})(\partial G(\dot{\mathbf{U}}_n) - \partial G(\dot{\mathbf{U}}))\|_2^2 = \int_{\Omega} Y(\mathbf{p})^2 (\partial G(\dot{\mathbf{U}}_n) - \partial G(\dot{\mathbf{U}}))^2 \, dx \rightarrow 0.$$

Indeed for every subsequence there exists a further subsequence such that $\dot{\mathbf{U}}_n \rightarrow \dot{\mathbf{U}}$ in L^2 and we can apply the dominated convergence theorem to pass to the limit. $\square$

Lemma 2.10. The operator $\mathcal{T}_p : \mathcal{X} \rightarrow \mathcal{X}'$ is coercive uniformly in $\mathbf{p}$.

Proof. For any $\xi \in \mathcal{X}$ we have

$$\begin{aligned} \langle \mathcal{T}_p(\xi), \xi \rangle &= \int_{\Omega} \tau \mathfrak{E}(\mathbf{E}\dot{\mathbf{u}} - \dot{\mathbf{U}}) : (\mathbf{E}\dot{\mathbf{u}} - \dot{\mathbf{U}}) \, dx + \int_{\Omega} \mathfrak{D}\mathbf{E}\dot{\mathbf{u}} : \mathbf{E}\dot{\mathbf{u}} \, dx + \\ &+ \int_{\Omega} (\mathfrak{E} + \tau \mathfrak{A}) \, \text{grad}^2 \dot{\mathbf{u}} : \text{grad}^2 \dot{\mathbf{u}} \, dx + \int_{\Omega} [(\mathfrak{F} + \tau \mathfrak{B})\dot{\mathbf{U}} \circ \dot{\mathbf{U}} + Y(\mathbf{p})\partial G(\dot{\mathbf{U}}) \circ \dot{\mathbf{U}}] \, dx. \end{aligned}$$

Using the hypotheses (E, b) and (G, b) and using that $\dot{\mathbf{U}} : \mathbf{I} = 0$ since $\dot{\mathbf{U}} \in \text{Dev}$ we have

$$\langle \mathcal{T}_p(\xi), \xi \rangle \geq \tau\gamma \|\mathbf{E}\dot{\mathbf{u}} - \dot{\mathbf{U}}\|_2^2 + \tau\alpha \|\text{grad } \mathbf{E}\dot{\mathbf{u}}\|_2^2 + (\phi + \tau\beta) \|\dot{\mathbf{U}}\|_2^2.$$

By the Young inequality

$$\|\mathbf{E}\dot{\mathbf{u}} - \dot{\mathbf{U}}\|_2^2 \geq (1 - \theta) \|\mathbf{E}\dot{\mathbf{u}}\|_2^2 + \left(1 - \frac{1}{\theta}\right) \|\dot{\mathbf{U}}\|_2^2$$

for any $\theta > 0$. Then

$$\begin{aligned} \langle \mathcal{T}_p(\xi), \xi \rangle &\geq (1 - \theta) \tau\gamma \|\mathbf{E}\dot{\mathbf{u}}\|_2^2 + \tau\alpha \|\text{grad } \mathbf{E}\dot{\mathbf{u}}\|_2^2 + \\ &\quad + \left(1 - \frac{1}{\theta}\right) \tau\gamma \|\dot{\mathbf{U}}\|_2^2 + (\phi + \tau\beta) \|\dot{\mathbf{U}}\|_2^2. \end{aligned}$$

We choose $\theta < 1$ sufficiently close to 1 such there exists a constant $C_1 > 0$ with the property that

$$\left(1 - \frac{1}{\theta}\right) \tau\gamma \|\dot{\mathbf{U}}\|_2^2 + (\phi + \tau\beta) \|\dot{\mathbf{U}}\|_2^2 \geq C_1 \|\dot{\mathbf{U}}\|_2^2 \geq C_2 \|\dot{\mathbf{U}}\|_{1,2}^2.$$

The thesis follows by application of Korn inequality. $\square$

Lemma 2.11. The operator $\mathcal{T}_p : \mathcal{X} \rightarrow \mathcal{X}'$ is strongly monotone uniformly in p , i.e., there exists $M > 0$ such that

$$\langle \mathcal{T}_p(\xi_2) - \mathcal{T}_p(\xi_1), \xi_2 - \xi_1 \rangle \geq M \|\xi_2 - \xi_1\|_{\mathcal{X}}^2$$

for all $p \in \mathcal{Q}$ and ξ_1, ξ_2 in $\mathcal{X}$.

Proof. For all ξ_1 and ξ_2 in $\mathcal{X}$ we have

$$\begin{aligned} &\langle \mathcal{T}_p(\xi_2) - \mathcal{T}_p(\xi_1), \xi_2 - \xi_1 \rangle = \\ &= \int_{\Omega} \tau\mathfrak{C}(\mathbf{E}(\dot{\mathbf{u}}_2 - \dot{\mathbf{u}}_1) - (\dot{\mathbf{U}}_2 - \dot{\mathbf{U}}_1)) : (\mathbf{E}(\dot{\mathbf{u}}_2 - \dot{\mathbf{u}}_1) - (\mathbf{U}_2 - \mathbf{U}_1)) \, dx + \\ &\quad + \int_{\Omega} \mathfrak{D}(\mathbf{E}(\dot{\mathbf{u}}_2 - \dot{\mathbf{u}}_1)) : (\mathbf{E}(\dot{\mathbf{u}}_2 - \dot{\mathbf{u}}_1)) \, dx + \\ &\quad + \int_{\Omega} (\mathfrak{E} + \tau\mathfrak{A})(\text{grad}^2 \dot{\mathbf{u}}_2 - \text{grad}^2 \dot{\mathbf{u}}_1) : (\text{grad}^2 \dot{\mathbf{u}}_2 - \text{grad}^2 \dot{\mathbf{u}}_1) \, dx + \\ &\quad + \int_{\Omega} [(\mathfrak{F} + \tau\mathfrak{B})(\dot{\mathbf{U}}_2 - \dot{\mathbf{U}}_1) \circ (\dot{\mathbf{U}}_2 - \dot{\mathbf{U}}_1) + \mathbf{Y}(p)(\partial G(\dot{\mathbf{U}}_2) - \partial G(\dot{\mathbf{U}}_1)) : (\dot{\mathbf{U}}_2 - \dot{\mathbf{U}}_1)] \, dx \geq \\ &\quad \geq (\delta + \tau\gamma) \|\mathbf{E}(\dot{\mathbf{u}}_2 - \dot{\mathbf{u}}_1) - (\dot{\mathbf{U}}_2 - \dot{\mathbf{U}}_1)\|_2^2 + (\phi + \tau\beta) \|\dot{\mathbf{U}}_2 - \dot{\mathbf{U}}_1\|_2^2. \end{aligned}$$

We have used that since G is convex then ∂G is monotone. Reasoning as in the proof of Lemma 2.10 we have

$$\begin{aligned} \langle \mathcal{T}_p(\xi_2) - \mathcal{T}_p(\xi_1), \xi_2 - \xi_1 \rangle &\geq (1 - \theta)(\delta + \tau\gamma) \|\mathbf{E}(\dot{\mathbf{u}}_2 - \dot{\mathbf{u}}_1)\|_2^2 + (\alpha + \tau\epsilon) \|\text{grad } \mathbf{E}(\dot{\mathbf{u}}_2 - \dot{\mathbf{u}}_1)\|_2^2 + \\ &\quad + \left(1 - \frac{1}{\theta}\right) (\delta + \tau\gamma) \|\dot{\mathbf{U}}_2 - \dot{\mathbf{U}}_1\|_2^2 + \\ &\quad + (\phi + \tau\beta) \|\dot{\mathbf{U}}_2 - \dot{\mathbf{U}}_1\|_2^2. \end{aligned}$$

Choosing $0 < \theta < 1$ sufficiently close to 1 and using second order Korn inequality we conclude. $\square$

Now applying Theorem C.16 for each $p \in \mathcal{Q}$ there exists $\xi \in \mathcal{X}$ such that

$$\langle \mathcal{T}_p(\xi), \tilde{\zeta} \rangle = \langle l, \mathcal{J}(\tilde{\zeta}) \rangle \quad \text{for all } \tilde{\zeta} \in \mathcal{X}.$$

Moreover ξ is unique. Indeed by strong monotonicity if ξ_1 and ξ_2 are two solutions we have

$$M \|\xi_2 - \xi_1\|_{\mathcal{X}}^2 \leq \langle \mathcal{T}_p(\xi_2) - \mathcal{T}_p(\xi_1), \xi_2 - \xi_1 \rangle = 0.$$

We call $\Xi: \mathcal{Q} \rightarrow \mathcal{X}$ the map that assign to $p \in \mathcal{Q}$ the unique ξ . Now to conclude the existence of a solution to Problem 2.8 we need to solve in $p \in \mathcal{Q}$ the equation

$$\langle \mathcal{A}(p, \mathcal{J}(\Xi(p))), (0, 0, \tilde{\mu}) \rangle + \langle Bp, (0, 0, \tilde{\mu}) \rangle = \langle l, (0, 0, \tilde{\mu}) \rangle \quad \text{for all } \tilde{\mu} \in L^2(\Omega)$$

that is

$$\int_{\Omega} \text{tr} [\tau \mathfrak{C} (E\dot{u} - \dot{U})] \frac{\tilde{\mu}}{d} dx - \int_{\Omega} p \tilde{\mu} dx = \int_{\Omega} \text{tr} [\mathfrak{C} (Eu - U)] \frac{\tilde{\mu}}{d} dx$$

for all $\tilde{\mu} \in L^2(\Omega)$, where $\dot{y} = (\dot{u}, \dot{U}, \dot{\lambda}) = \mathcal{J}(\Xi(p))$. This is equivalent to

$$p = \frac{1}{d} \text{tr} [\tau \mathfrak{C} (E\dot{u} - \dot{U})] - \frac{1}{d} \text{tr} [\mathfrak{C} (Eu - U)] = \mathcal{S}(p) \quad (2.2.5)$$

where we have introduced the operator $\mathcal{S}: \mathcal{Q} \rightarrow \mathcal{Q}$. We want to prove the existence of a fixed point of $\mathcal{S}$ using the Schauder fixed point theorem (Theorem C.11). To prove that $\mathcal{S}$ satisfy the required hypotheses we need to establish some regularity results regarding Ξ . We first need the following lemma.

Lemma 2.12. The operator $\mathcal{A}$ is Lipschitz continuous in p uniformly in $\dot{y}$.

Proof. By (Y, a) and (G, c) we have

$$\begin{aligned} |\langle \mathcal{A}(p, \dot{y}) - \mathcal{A}(q, \dot{y}), \tilde{z} \rangle| &\leq \int_{\Omega} |Y(p) - Y(q)| |\partial G(\dot{U})| |\tilde{V}| dx \leq \\ &\leq \text{Lip}(Y) \|\partial G\|_{\infty} \int_{\Omega} |p - q| |\tilde{V}| dx \leq L \|p - q\|_2 \|\tilde{V}\|_2 \end{aligned}$$

for some constant $L > 0$. Then we have

$$\|\mathcal{A}(p, \dot{y}) - \mathcal{A}(q, \dot{y})\|_{\mathcal{Z}'} \leq L \|p - q\|_{\mathcal{Q}}$$

This concludes the proof. □

Now we can prove the continuity of Ξ .

Lemma 2.13. The operator Ξ is continuous and has bounded range.

Proof. First we prove that Ξ has bounded range. We assume by contradiction that there exists $p_n \in \mathcal{Q}$ such that $\|\Xi(p_n)\|_{\mathcal{X}} \geq n$. Let $\xi_n = \Xi(p_n)$. Now

$$\frac{\langle \mathcal{T}_{p_n}(\xi_n), \xi_n \rangle}{\|\xi_n\|_{\mathcal{X}}} \leq \|\langle l, \mathcal{J}(\cdot) \rangle\|_{\mathcal{X}'}$$

But by Lemma 2.10

$$\frac{\langle \mathcal{T}_p(\xi_n), \xi_n \rangle}{\|\xi_n\|_{\mathcal{X}}} \rightarrow \infty$$

and this is a contradiction. Now we prove that Ξ is Lipschitz continuous. Indeed let $p_i \in \mathcal{Q}$ and $\xi_i = \Xi(p_i)$ with i equals to 1 or 2. Since $\mathcal{T}_{p_1}(\xi_1) = \mathcal{T}_{p_2}(\xi_2)$, $\mathcal{T}_p$ is strictly monotone uniformly in p and $\mathcal{A}$ is uniformly Lipschitz in p we have

$$\begin{aligned} M\|\xi_2 - \xi_1\|^2 &\leq \langle \mathcal{T}_{p_1}(\xi_2) - \mathcal{T}_{p_1}(\xi_1), \xi_2 - \xi_1 \rangle = \langle \mathcal{T}_{p_1}(\xi_2) - \mathcal{T}_{p_2}(\xi_2), \xi_2 - \xi_1 \rangle \leq \\ &\leq L\|p_2 - p_1\|_{\mathcal{Q}}\|\xi_2 - \xi_1\|_{\mathcal{X}}. \end{aligned}$$

This concludes the proof. $\square$

The following lemma establish the regularity of $\mathcal{J}$.

Lemma 2.14. The map $\mathcal{J}: \mathcal{X} \rightarrow \mathcal{Z}$ is continuous and bounded.

Proof. It is trivial since $\mathcal{J}(\xi) = (\dot{u}, \dot{U}, 0)$. $\square$

Now we are ready to prove the desired properties of $\mathcal{S}$.

Lemma 2.15. The map $\mathcal{S}: \mathcal{Q} \rightarrow \mathcal{Q}$ is continuous and has a bounded and compact range.

Proof. We notice that since Ξ is continuous and has compact range then also $\mathcal{S}$ is continuous and has compact range. We notice that by Rellich–Kondarchov theorem then $\dot{u} \mapsto E\dot{u}$ from H^2 to L^2 and $\dot{U} \mapsto \dot{U}$ from H^1 to L^2 are compact. Then $\mathcal{S}$ is compact. Since $\mathcal{S}$ has bounded range and is compact then it has compact range. $\square$

Now we are in position to apply Schauder fixed point theorem and we get the existence of a fixed point of $\mathcal{S}$. We have obtained the desired result.

Proposition 2.16. Problem 2.8 admits at least a solution.

Now we redefine the operator $\mathcal{A}: \mathcal{Z} \rightarrow \mathcal{Z}'$ by

$$\langle \mathcal{A}y, \tilde{z} \rangle = \int_{\Omega} \mathfrak{C} \left(Eu - u - \frac{\lambda}{d} I \right) : \left(E\tilde{v} - \tilde{V} - \frac{\tilde{\mu}}{d} I \right) dx + \int_{\Omega} \mathfrak{A} \text{grad } u : \text{grad } \tilde{v} dx + \int_{\Omega} \mathfrak{B} U \circ \tilde{V} dx$$

and we define operator $\mathcal{E}: \mathcal{Z} \rightarrow \mathcal{Z}'$ by

$$\langle \mathcal{E}y, \tilde{z} \rangle = \int_{\Omega} \mathfrak{D} E\dot{u} : E\tilde{v} dx + \int_{\Omega} \mathfrak{C} \text{grad } \dot{u} : \text{grad } \tilde{v} dx + \int_{\Omega} \mathfrak{F} \dot{U} \circ \tilde{V} dx.$$

We define also $\mathcal{D}: \mathcal{Q} \times \mathcal{Z} \rightarrow \mathcal{Z}'$ by

$$\langle \mathcal{D}(p, \dot{y}), \tilde{z} \rangle = \int_{\Omega} Y(p) \partial G(\dot{U}) : \tilde{V} dx.$$

The operator $\mathcal{B}$ is defined exactly as before and $\mathcal{C}: \mathcal{Z} \rightarrow \mathcal{Q}'$ is redefined by

$$\langle \mathcal{C}y, \tilde{q} \rangle = \int_{\Omega} \lambda \tilde{q} dx = 0.$$

Let also $l: [0, T] \rightarrow \mathcal{Z}'$ redefined by

$$\langle l(t), \tilde{z} \rangle = \int_{\Omega} f(t) \cdot \tilde{v} dx + \int_{\Gamma_N} h(t) \cdot \tilde{v} dx.$$

Proposition 2.16 ensures that for each N there exists $\{(\dot{y}_n^N, p_n^N, y_n^N)\}_{n=0}^N$ such that $(\dot{y}_0^N, p_0^N, y_0^N) = 0$, $y_n^N = y_{n-1}^N + \tau \dot{y}_n^N$ and

$$\begin{cases} \mathcal{A}y_n^N + \mathcal{E}\dot{y}_n^N + \mathcal{D}(p_n^N, \dot{y}_n^N) + \mathcal{B}p_n^N = l_n^N \\ \mathcal{C}y_n^N = 0 \end{cases} \quad (2.2.6)$$

for all $n \in \{1, \dots, N\}$. Here $l_n^N = l(t_n^N)$.

Construction of the continuous in time solution

Now we want to construct the continuous in time solution passing to the limit $N \rightarrow \infty$. We define

$$\begin{aligned} l^N &\in L^2((0, T), \mathcal{Z}') && \text{piecewise constant interpolation of } \{l_n^N\}_{n=1}^N \\ \bar{y}^N &\in L^2((0, T), \mathcal{Z}) && \text{piecewise constant interpolation of } \{y_n^N\}_{n=1}^N \\ y^N &\in H^1((0, T), \mathcal{Z}) && \text{piecewise linear interpolation of } \{y_n^N\}_{n=1}^N \\ \dot{y}^N &\in L^2((0, T), \mathcal{Z}) && \text{piecewise constant interpolation of } \{\dot{y}_n^N\}_{n=1}^N \\ p^N &\in L^2((0, T), \mathcal{Q}) && \text{piecewise constant interpolation of } \{p_n^N\}_{n=1}^N. \end{aligned}$$

We will need the following lemma.

Lemma 2.17. We have that $l^N \rightarrow l$ in $L^2((0, T), \mathcal{Z}')$ and in particular

$$\|l^N\|_{L^2((0, T), \mathcal{Z}')}^2 = \sum_{n=1}^N \tau_N \|l_n^N\|_{\mathcal{Z}'}^2$$

is bounded.

Proof. For all $t \in [t_{n-1}^N, t_n^N]$ we have

$$l^N(t) = l_n^N = l(t_n^N) = l(t) + \int_t^{t_n^N} \dot{l}(s) ds.$$

By Hölder inequality

$$\|l^N(t) - l(t)\|_{\mathcal{Z}'} \leq \int_t^{t_n^N} \|\dot{l}(s)\|_{\mathcal{Z}'} ds \leq \int_{t_{n-1}^N}^{t_n^N} \|\dot{l}(s)\|_{\mathcal{Z}'} ds \leq \sqrt{\tau_N} \left[\int_{t_{n-1}^N}^{t_n^N} \|\dot{l}(s)\|_{\mathcal{Z}'}^2 ds \right]^{1/2}.$$

Taking the square and integrating for $t \in [t_{n-1}^N, t_n^N]$ we have

$$\int_{t_{n-1}^N}^{t_n^N} \|l^N(t) - l(t)\|_{\mathcal{Z}'}^2 dt \leq \tau_N^2 \int_{t_{n-1}^N}^{t_n^N} \|\dot{l}(s)\|_{\mathcal{Z}'}^2 ds.$$

Adding this inequalities for $n \in \{1, \dots, N\}$ we have

$$\|l^N - l\|_{L^2((0, T), \mathcal{Z}')}^2 \leq \tau_N^2 \|\dot{l}\|_{L^2((0, T), \mathcal{Z}')}^2 \rightarrow 0.$$

This concludes the proof. $\square$

We now obtain a priori estimates on $\bar{y}^N$, y^N and $\dot{y}^N$.

Lemma 2.18. The functions $\bar{y}^N$ and y^N are bounded in $L^\infty((0, T), \mathcal{Z})$ and the function $\dot{y}^N$ is bounded in $L^2((0, T), \mathcal{Z})$.

Proof. We first observe that y_n^N and $\dot{y}_n^N$ belongs to $\mathcal{N}$ and that $\mathcal{A}$ and $\mathcal{E}$ are coercive there. Next we test the first equation in (2.2.6) with $\tilde{z} = \dot{y}_n^N$. We notice that $\langle \mathcal{D}(p_n^N, \dot{y}_n^N), \dot{y}_n^N \rangle \geq 0$ and $\langle \mathcal{B}p_n^N, \dot{y}_n^N \rangle = 0$. Then

$$\frac{1}{\tau_N} \langle \mathcal{A}y_n^N, y_n^N - y_{n-1}^N \rangle + \langle \mathcal{E}\dot{y}_n^N, \dot{y}_n^N \rangle \leq \langle l_n^N, \dot{y}_n^N \rangle.$$

Multiplying by $2\tau_N$ we have

$$2\langle \mathcal{A}y_n^N, y_n^N - y_{n-1}^N \rangle + 2\tau_N \langle \varepsilon \dot{y}_n^N, \dot{y}_n^N \rangle \leq 2\tau_N \langle l_n^N, \dot{y}_n^N \rangle.$$

Now

$$\begin{aligned} 2\langle \mathcal{A}y_n^N, y_n^N - y_{n-1}^N \rangle &= \langle \mathcal{A}y_n^N, y_n^N \rangle - \langle \mathcal{A}y_{n-1}^N, y_{n-1}^N \rangle + \langle \mathcal{A}(y_n^N - y_{n-1}^N), (y_n^N - y_{n-1}^N) \rangle \geq \\ &\geq \langle \mathcal{A}y_n^N, y_n^N \rangle - \langle \mathcal{A}y_{n-1}^N, y_{n-1}^N \rangle. \end{aligned}$$

Let $E > 0$ be its coercivity constant of ε in $\mathcal{N}$. Then

$$2\tau_N \langle \varepsilon \dot{y}_n^N, \dot{y}_n^N \rangle \geq 2\tau_N E \|\dot{y}_n^N\|_{\mathcal{Z}}^2$$

and using Young inequality

$$2\tau_N \langle l_n^N, \dot{y}_n^N \rangle \leq \frac{\tau_N}{E} \|l_n^N\|_{\mathcal{Z}'}^2 + \tau_N E \|\dot{y}_n^N\|_{\mathcal{Z}}^2.$$

Then

$$\langle \mathcal{A}y_n^N, y_n^N \rangle - \langle \mathcal{A}y_{n-1}^N, y_{n-1}^N \rangle + \tau_N E \|\dot{y}_n^N\|_{\mathcal{Z}}^2 \leq \frac{\tau_N}{E} \|l_n^N\|_{\mathcal{Z}'}^2. \quad (2.2.7)$$

Let $A > 0$ the coercivity constant of $\mathcal{A}$ on $\mathcal{N}$. Then adding the inequalities (2.2.7) we have

$$A \|y_n^N\|_{\mathcal{Z}}^2 + E \sum_{k=1}^n \tau_N \|\dot{y}_k^N\|_{\mathcal{Z}}^2 \leq \langle \mathcal{A}y_n^N, y_n^N \rangle + E \sum_{k=1}^n \tau_N \|\dot{y}_k^N\|_{\mathcal{Z}}^2 \leq \frac{1}{E} \sum_{k=1}^n \tau_N \|l_k^N\|_{\mathcal{Z}'}^2.$$

Using Lemma 2.17 we have that

$$\|y_n^N\|_{\mathcal{Z}} \quad \text{and} \quad \sum_{n=1}^N \tau_N \|\dot{y}_n^N\|_{\mathcal{Z}}^2$$

are bounded uniformly in $N \in \mathbb{N}$ and $n \in \{0, \dots, N\}$. This implies immediately the thesis. $\square$

Using that lemma, up to extracting a subsequence we have that

$$\bar{y}^N \xrightarrow{*} \bar{y} \quad \text{in } L^\infty((0, T), \mathcal{Z}) \quad (2.2.8a)$$

$$y^N \xrightarrow{*} y \quad \text{in } L^\infty((0, T), \mathcal{Z}) \quad (2.2.8b)$$

$$\dot{y}^N \rightharpoonup z \quad \text{in } L^2((0, T), \mathcal{Z}). \quad (2.2.8c)$$

We notice that from $\lambda_n^N = 0$ it follows immediately that $\lambda = 0$.

Lemma 2.19. We have that $\bar{y} = y \in H^1((0, T), \mathcal{Z})$ and $\dot{y} = z$. Moreover $y(0) = 0$.

Proof. We prove that $\bar{y}^N - y^N \rightarrow 0$ in $L^\infty((0, T), \mathcal{Z})$. Indeed let $t \in [t_{n-1}^N, t_n^N]$. Then

$$\|\bar{y}^N(t_n^N) - y(t_n^N)\|_{\mathcal{Z}} = \|y_n^N - y_{n-1}^N - (t_n^N - t) \dot{y}_n^N\|_{\mathcal{Z}} \leq \tau_N \|\dot{y}_n^N\|_{\mathcal{Z}}.$$

It follows that

$$\|\bar{y}^N(t_n^N) - y(t_n^N)\|_{\mathcal{Z}}^2 \leq \tau_N^2 \|\dot{y}_n^N\|_{\mathcal{Z}}^2 \leq \tau_N \|\dot{y}^N\|_{L^2((0, T), \mathcal{Z})}^2 \rightarrow 0.$$

This establish the equality $\bar{y} = y$. Now let $\phi \in C_c^\infty([0, T])$ and $\zeta \in \mathcal{Z}'$ arbitrary. Then

$$\int_0^T \phi \langle \zeta, \dot{y}^N \rangle dt = -\phi(0) \langle \zeta, y^N(0) \rangle - \int_0^T \dot{\phi} \langle \zeta, y^N \rangle dt.$$

Choosing first $\phi \in C_c^\infty((0, T))$ and exploiting (2.2.8) we have

$$\int_0^T \phi \langle \zeta, \dot{y} \rangle dt = - \int_0^T \dot{\phi} \langle \zeta, y \rangle dt.$$

This means exactly that $y \in H^1((0, T), \mathcal{Z})$ and $\dot{y} = z$. In particular for $\phi \in C_c^\infty([0, T])$ generic we have

$$\int_0^T \phi \langle \zeta, \dot{y} \rangle dt = -\phi(0) \langle \zeta, y(0) \rangle - \int_0^T \dot{\phi} \langle \zeta, y \rangle dt.$$

Then we get also

$$\langle \zeta, y^N(0) \rangle \rightarrow \langle \zeta, y(0) \rangle.$$

Since $y^N(0) = 0$ also $y(0) = 0$. □

We now discuss the convergence of p^N .

Lemma 2.20. Up to extracting a further subsequence $u^N \rightarrow u$ in $C([0, T], H^1(\Omega)^d)$ and $U^N \rightarrow U$ in $C([0, T], L^2(\Omega, \text{DevSym}))$. Moreover $p^N \rightarrow p$ in $C([0, T], \mathcal{Q})$ for some p .

Proof. Since $y^N \rightharpoonup y$ in $L^\infty((0, T), \mathcal{Z})$ and $\dot{y}^N \rightharpoonup \dot{y}$ in $L^2((0, T), \mathcal{Z})$ then also

$$\begin{aligned} u^N &\rightharpoonup u \text{ in } L^\infty(\Omega, H^2) \quad \text{and} \quad \dot{u}^N \rightharpoonup \dot{u} \text{ in } L^2(\Omega, H^2), \\ U^N &\rightharpoonup U \text{ in } L^\infty(\Omega, H^1) \quad \text{and} \quad \dot{U}^N \rightharpoonup \dot{U} \text{ in } L^2(\Omega, H^1). \end{aligned}$$

By Aubin–Lions–Simon lemma (Theorem C.2)

$$u^N \rightarrow u \text{ in } C([0, T], H^1) \quad \text{and} \quad U^N \rightarrow U \text{ in } C([0, T], L^2).$$

We recall that by (2.2.5) we have

$$p_n^N = \frac{1}{d} \text{tr} [\tau_N \mathfrak{C} (E \dot{u}_n^N - \dot{U}_n^N)] - \frac{1}{d} \text{tr} [\mathfrak{C} (E u_n^N - U_n^N)]. \quad (2.2.9)$$

Since $\tau_N \rightarrow 0$ and $\dot{y}_n^N$ is bounded in $L^2((0, T), \mathcal{Z})$ we have

$$p^N \rightarrow p = -\frac{1}{d} \text{tr} [\mathfrak{C} (E u - U)] \quad \text{in } L^2((0, T), \mathcal{Z}).$$

This concludes the proof. □

Remark 2.21. The fact that $p \in C^0([0, T], \mathcal{Q})$ suggest that there are not bifurcation modes of pressure.

Now we notice that (2.2.6) implies that for all $\tilde{z} \in L^2((0, T), \mathcal{Z})$ we have

$$\int_0^T \langle \mathcal{A} \bar{y}^N, \tilde{z} \rangle dt + \int_0^T \langle \mathcal{E} \dot{y}^N, \tilde{z} \rangle dt + \int_0^T \langle \mathcal{D}(P^N, \dot{y}^N), \tilde{z} \rangle dt + \int_0^T \langle \mathcal{B} p^N, \tilde{z} \rangle dt = \int_0^T \langle l^N, \tilde{z} \rangle dt.$$

By Lemma 2.17 and (2.2.8) we have

$$\begin{aligned} \int_0^T \langle l^N, \tilde{z} \rangle dt &\rightarrow \int_0^T \langle l, \tilde{z} \rangle dt \\ \int_0^T \langle \mathcal{A} \bar{y}^N, \tilde{z} \rangle dt &\rightarrow \int_0^T \langle \mathcal{A} y, \tilde{z} \rangle dt \\ \int_0^T \langle \mathcal{E} \dot{y}^N, \tilde{z} \rangle dt &\rightarrow \int_0^T \langle \mathcal{E} \dot{y}, \tilde{z} \rangle dt \\ \int_0^T \langle \mathcal{B} p^N, \tilde{z} \rangle dt &\rightarrow \int_0^T \langle \mathcal{B} p, \tilde{z} \rangle dt. \end{aligned}$$

Moreover we have

$$\begin{aligned} \int_0^T \|\mathcal{D}(p^N, \dot{y}^N)\|_{\mathcal{Z}'}^2 dt &\leq \int_0^T \int_{\Omega} Y(p^N)^2 |\partial G(\dot{U}^N)|^2 dx dt \leq \\ &\leq \int_0^T \int_{\Omega} (Y(0) + \text{Lip}(Y)|p^N|)^2 \|\partial G\|_{\infty}^2 dx dt \lesssim \\ &\lesssim (1 + \|p^N\|_{L^2((0,T),\mathcal{Q})}^2). \end{aligned}$$

But since p^N converges in $C([0, T], \mathcal{Q})$ then $\mathcal{D}(p^N, \dot{y}^N)$ is bounded in $L^2((0, T), \mathcal{Z}')$. Then up to extracting a further subsequence

$$\mathcal{D}(p^N, \dot{y}^N) \rightharpoonup \delta \text{ in } L^2((0, T), \mathcal{Z}')$$

for some $\delta \in L^2((0, T), \mathcal{Z}')$. Then we have

$$\int_0^T \langle \mathcal{A}y, \tilde{z} \rangle dt + \int_0^T \langle \mathcal{E}\dot{y}, \tilde{z} \rangle dt + \int_0^T \langle \delta, \tilde{z} \rangle dt + \int_0^T \langle \mathcal{B}p, \tilde{z} \rangle dt = \int_0^T \langle l, \tilde{z} \rangle dt \quad (2.2.10)$$

for all $\tilde{z} \in L^2((0, T), \mathcal{Z})$. This implies also

$$\langle \mathcal{A}y, \tilde{z} \rangle + \langle \mathcal{E}\dot{y}, \tilde{z} \rangle + \langle \delta, \tilde{z} \rangle + \langle \mathcal{B}p, \tilde{z} \rangle = \langle l, \tilde{z} \rangle$$

for almost every $t \in (0, T)$. If we prove that $\delta = \mathcal{D}(p, \dot{y})$ we have that (y, p) is a global in time solution to Problem 2.6 and we have concluded. To this aim let us define $\mathcal{Z}_T = L^2((0, T), \mathcal{Z})$ and $\mathcal{Q}_T = L^2((0, T), \mathcal{Q})$. Let $\mathcal{J}: \mathcal{Q} \times \mathcal{Z} \rightarrow \mathbb{R}$ be defined by

$$\mathcal{J}(p, \dot{y}) = \int_{\Omega} Y(p)G(\dot{U})dx.$$

Let also $\mathcal{J}_T: \mathcal{Q}_T \times \mathcal{Z}_T \rightarrow \mathbb{R}$ and $\mathcal{D}_T: \mathcal{Q}_T \times \mathcal{Z}_T \rightarrow \mathcal{Z}'_T$ defined by

$$\mathcal{J}_T(p, \dot{y}) = \int_0^T \mathcal{J}(p, \dot{y})dt \quad \text{and} \quad \langle \mathcal{D}_T(p, \dot{y}), \tilde{z} \rangle = \int_0^T \langle \mathcal{D}(p, \dot{y}), \tilde{z} \rangle dt$$

for all $\tilde{z} \in L^2((0, T), \mathcal{Z})$. We notice that for all $y \in \mathcal{Z}_T$ and $p \in \mathcal{Q}_T$ we can identify $\mathcal{D}_T(p, \dot{y}) \in \mathcal{Z}'_T$ with $\mathcal{D}(p, \dot{y}) \in L^2((0, T), \mathcal{Z}')$.

Lemma 2.22. Let $\dot{y} \in L^2((0, T), \mathcal{Z})$, $p \in L^2((0, T), \mathcal{Q})$ and $\delta \in L^2((0, T), \mathcal{Z}')$. Then $\delta = \mathcal{D}(p, \dot{y})$ if and only if

$$\mathcal{J}_T(p, \tilde{z}) \geq \mathcal{J}_T(p, \dot{y}) + \int_0^T \langle \delta, \tilde{z} - \dot{y} \rangle dt \quad (2.2.11)$$

for all $\tilde{z} \in L^2((0, T), \mathcal{Z})$.

Proof. It is easy to see that $\mathcal{J}$ is convex in $\dot{y}$ and is Gateaux differentiable in $\dot{y}$ with Gateaux differential

$$\langle D_{\dot{y}}\mathcal{J}_T(p, \dot{y}), \tilde{z} \rangle = \int_0^T \langle \mathcal{D}(p, \dot{y}), \tilde{z} \rangle dt = \langle \mathcal{D}_T(p, \dot{y}), \tilde{z} \rangle$$

for all $\tilde{z} \in L^2((0, T), \mathcal{Z})$. By [13, Proposition 5.3] then $\mathcal{J}_T$ is sub differentiable and $D_{\dot{y}}^{\text{sub}}\mathcal{J}_T(p, \dot{y}) = \{D_{\dot{y}}\mathcal{J}_T(p, \dot{y})\}$. Then $\delta = \mathcal{D}(p, \dot{y})$ if and only if $\delta \in D_{\dot{y}}^{\text{sub}}\mathcal{J}_T(p, \dot{y})$ but this last inclusion is equivalent to (2.2.11). $\square$

By that lemma we have

$$\mathcal{J}_T(\mathbf{p}^N, \tilde{\mathbf{z}}) \geq \mathcal{J}_T(\mathbf{p}^N, \dot{\mathbf{y}}) + \langle \mathcal{D}_T(\mathbf{p}^N, \dot{\mathbf{y}}^N), \tilde{\mathbf{z}} - \dot{\mathbf{y}}^N \rangle. \quad (2.2.12)$$

Now we want to pass to the limit in this inequality. We will need the following lemma.

Lemma 2.23. For each $\dot{\mathbf{y}} \in \mathcal{Z}_T$ the functional $\mathcal{J}_T(\cdot, \dot{\mathbf{y}})$ is Lipschitz continuous with Lipschitz constant bounded by $\|\dot{\mathbf{y}}\|_{\mathcal{Z}_T}$. Moreover for each $\mathbf{p} \in \mathcal{Q}_T$ we have that $\mathcal{J}_T(\mathbf{p}, \cdot)$ is convex and continuous. In particular $\mathcal{J}_T(\mathbf{p}, \cdot)$ is weakly lower semicontinuous.

Proof. We have that

$$\begin{aligned} |\mathcal{J}_T(\mathbf{p}_2, \dot{\mathbf{y}}) - \mathcal{J}_T(\mathbf{p}_1, \dot{\mathbf{y}})| &\leq \int_0^T \int_{\Omega} |Y(\mathbf{p}_2) - Y(\mathbf{p}_1)| |G(\dot{\mathbf{U}})| dx \leq \\ &\leq \|\partial G\|_{\infty} \text{Lip}(Y) \int_0^T \int_{\Omega} |\mathbf{p}_2 - \mathbf{p}_1| |\dot{\mathbf{U}}| dt dx \lesssim \\ &\lesssim \|\mathbf{p}_2 - \mathbf{p}_1\|_{\mathcal{Q}_T} \|\dot{\mathbf{y}}\|_{\mathcal{Z}_T}. \end{aligned}$$

This proves that for each $\dot{\mathbf{y}} \in \mathcal{Z}_T$ the functional $\mathcal{J}_T(\cdot, \dot{\mathbf{y}})$ is Lipschitz continuous with Lipschitz constant bounded by $\|\dot{\mathbf{y}}\|_{\mathcal{Z}_T}$. The convexity of $\mathcal{J}_T(\mathbf{p}, \cdot)$ follows immediately from the convexity of G . Moreover

$$\begin{aligned} |\mathcal{J}_T(\mathbf{p}, \dot{\mathbf{y}}_2) - \mathcal{J}_T(\mathbf{p}, \dot{\mathbf{y}}_1)| &\leq \int_0^T \int_{\Omega} |Y(\mathbf{p})| |G(\dot{\mathbf{U}}_2) - G(\dot{\mathbf{U}}_1)| dx \leq \\ &\leq \|\partial G\|_{\infty} \int_0^T \int_{\Omega} (Y(0) + \text{Lip}(Y)|\mathbf{p}|) |\dot{\mathbf{U}}_2 - \dot{\mathbf{U}}_1| dt dx \lesssim \\ &\lesssim (1 + \|\mathbf{p}\|_{\mathcal{Q}_T}) \|\dot{\mathbf{y}}_2 - \dot{\mathbf{y}}_1\|_{\mathcal{Z}_T}. \end{aligned}$$

This proves the continuity of $\mathcal{J}_T(\mathbf{p}, \cdot)$ for each $\mathbf{p} \in \mathcal{Q}_T$ fixed. $\square$

Now we have

$$|\mathcal{J}_T(\mathbf{p}^N, \tilde{\mathbf{z}}) - \mathcal{J}_T(\mathbf{p}, \tilde{\mathbf{z}})| \leq C \|\mathbf{p}^N - \mathbf{p}\|_{\mathcal{Q}_T} \rightarrow 0$$

for some constant $C > 0$. Moreover

$$\mathcal{J}_T(\mathbf{p}^N, \dot{\mathbf{y}}^N) = [\mathcal{J}_T(\mathbf{p}^N, \dot{\mathbf{y}}^N) - \mathcal{J}_T(\mathbf{p}, \dot{\mathbf{y}}^N)] + \mathcal{J}_T(\mathbf{p}, \dot{\mathbf{y}}^N).$$

But since $\mathbf{p}^N \rightharpoonup$ in $\mathcal{Q}_T$ and $\dot{\mathbf{y}}^N$ are bounded in $\mathcal{Z}_T$

$$|\mathcal{J}_T(\mathbf{p}^N, \dot{\mathbf{y}}^N) - \mathcal{J}_T(\mathbf{p}, \dot{\mathbf{y}}^N)| \leq C \|\dot{\mathbf{y}}^N\|_{\mathcal{Z}_T} \|\mathbf{p}^N - \mathbf{p}\|_{\mathcal{Q}_T} \rightarrow 0$$

and

$$\liminf_{N \rightarrow \infty} \mathcal{J}_T(\mathbf{p}, \dot{\mathbf{y}}^N) \geq \mathcal{J}_T(\mathbf{p}, \dot{\mathbf{y}}).$$

Then

$$\liminf_{N \rightarrow \infty} \mathcal{J}_T(\mathbf{p}^N, \dot{\mathbf{y}}^N) \geq \mathcal{J}_T(\mathbf{p}, \dot{\mathbf{y}}).$$

Now by construction

$$\begin{aligned} \int_0^T \langle \mathcal{A} \dot{\mathbf{y}}^N, \tilde{\mathbf{z}} - \dot{\mathbf{y}}^N \rangle dt + \int_0^T \langle \mathcal{E} \dot{\mathbf{y}}^N, \tilde{\mathbf{z}} - \dot{\mathbf{y}}^N \rangle dt + \int_0^T \langle \mathcal{D}(\mathbf{p}^N, \dot{\mathbf{y}}^N), \tilde{\mathbf{z}} - \dot{\mathbf{y}}^N \rangle dt + \\ + \int_0^T \langle \mathcal{B} \mathbf{p}^N, \tilde{\mathbf{z}} - \dot{\mathbf{y}}^N \rangle dt = \int_0^T \langle \mathbf{l}^N, \tilde{\mathbf{z}} - \dot{\mathbf{y}}^N \rangle dt. \end{aligned}$$

But by Lemma 2.17, Lemma 2.20 and (2.2.8) we have

$$\begin{aligned} \int_0^T \langle \mathcal{A}y^N, \tilde{z} - \dot{y}^N \rangle dt &\rightarrow \int_0^T \langle \mathcal{A}y, \tilde{z} - \dot{y} \rangle dt \\ \int_0^T \langle \mathcal{B}p^N, \tilde{z} - \dot{y}^N \rangle dt &\rightarrow \int_0^T \langle \mathcal{B}p, \tilde{z} - \dot{y} \rangle dt \\ \int_0^T \langle l^N, \tilde{z} - \dot{y}^N \rangle dt &\rightarrow \int_0^T \langle l, \tilde{z} - \dot{y} \rangle dt \end{aligned}$$

since the convergence of y^N , p^N and l^N is strong. Moreover since

$$\dot{y} \rightarrow \int_0^T \langle \mathcal{E}\dot{y}^N, \dot{y}^N - \tilde{z} \rangle dt$$

is continuous and convex in $L^2((0, T), \mathcal{Z})$ it is weakly lower semicontinuous and

$$\liminf_{N \rightarrow \infty} \int_0^T \langle \mathcal{E}\dot{y}^N, \dot{y}^N - \tilde{z} \rangle dt \geq \int_0^T \langle \mathcal{E}\dot{y}, \dot{y} - \tilde{z} \rangle dt.$$

This implies that

$$\begin{aligned} \liminf_{N \rightarrow \infty} \langle \mathcal{D}_T(p^N, \dot{y}^N), \tilde{z} - \dot{y}^N \rangle &\geq \\ &\geq \int_0^T \langle l, \tilde{z} - \dot{y} \rangle dt - \int_0^T \langle \mathcal{A}y, \tilde{z} - \dot{y} \rangle dt - \int_0^T \langle \mathcal{E}\dot{y}, \tilde{z} - \dot{y} \rangle dt - \int_0^T \langle \mathcal{B}p, \tilde{z} - \dot{y} \rangle dt = \\ &= \int_0^T \langle \delta, \tilde{z} - \dot{y} \rangle dt \end{aligned}$$

where the last equality holds for (2.2.10). Collecting these results when we take the inferior limit for $N \rightarrow \infty$ of (2.2.12) and we get

$$\mathcal{J}_T(p, \tilde{z}) \geq \mathcal{J}_T(p, \dot{y}) + \int_0^T \langle \delta, \tilde{z} - \dot{y} \rangle dt.$$

Since now δ satisfy (2.2.11) we have by Lemma 2.2.11 that $\delta = \mathcal{D}(p, \dot{y})$ as desired.

2.2.5 Example of constitutive relations

In this section we give examples of constitutive relations satisfying the hypotheses of Section 2.2.1. Let $g: [0, \infty) \rightarrow \mathbb{R}$ be defined by

$$g(r) = \begin{cases} \frac{r^2}{4c} & \text{if } r \leq c \\ r - \frac{3}{4}c - \frac{c}{2} \log \frac{r}{c} & \text{if } r > c \end{cases}$$

where $c > 0$ so that

$$g'(r) = \begin{cases} \frac{r}{2c} & \text{if } r \leq c \\ 1 - \frac{c}{2r} & \text{if } r > c. \end{cases}$$

We set $G(\dot{U}) = g(|\dot{U}|)$. We let the yield strength be given by

$$Y(p) = \begin{cases} Y_0 & \text{if } p < -p_0 \\ Y_0 + \alpha(p + p_0) & \text{if } p \geq -p_0 \end{cases}$$

where $Y_0 \geq 0$, $p_0 \geq 0$ and $\alpha > 0$. We set $\mathfrak{A}$, $\mathfrak{B}$, $\mathfrak{D}$, $\mathfrak{E}$ and $\mathfrak{F}$ positive constants and $\mathfrak{C}$ and $\mathfrak{D}$ positive definite isotropic fourth order tensors.

2.2.6 Further regularity hypotheses

The local existence and uniqueness theorem is obtained under the following additional hypotheses.

$$(Y') \quad Y \in C^{1,1}(\mathbb{R})$$

$$(G') \quad G \in C^{2,1}(\text{DevSym})$$

2.2.7 Local existence and uniqueness and global uniqueness

Proposition 2.24. The map $\mathcal{F}$ is of class C^1 and the Fréchet derivative is given by

$$\begin{aligned} \langle D_{\dot{y}} \mathcal{F}(\dot{y}, p, y, t)z, (\tilde{z}, \tilde{q}) \rangle &= \int_{\Omega} \mathfrak{D} \text{Ev} : \tilde{E} \tilde{v} dx + \int_{\Omega} \mathfrak{E} \text{grad}^2 v : \text{grad}^2 \tilde{v} dx + \\ &\quad + \int_{\Omega} \mathfrak{F} \nabla \circ \tilde{\nabla} dx + \int_{\Omega} Y(p) \partial^2 G(\dot{U}) V : \tilde{V} dx \\ \langle D_p \mathcal{F}(\dot{y}, p, y, q)q, (\tilde{z}, \tilde{q}) \rangle &= \int_{\Omega} Y'(p) q \partial G(\dot{U}) : \tilde{V} dx - \int_{\Omega} q \tilde{\mu} dx \\ \langle D_y \mathcal{F}(\dot{y}, p, y, t)z, (\tilde{z}, \tilde{q}) \rangle &= \int_{\Omega} \mathfrak{E} \left(E v - V - \frac{\mu}{d} I \right) : \left(E \tilde{v} - \tilde{V} - \frac{\tilde{\mu}}{d} I \right) dx + \\ &\quad + \int_{\Omega} \mathfrak{A} \text{grad}^2 v : \text{grad}^2 \tilde{v} dx + \int_{\Omega} \mathfrak{B} \nabla \circ \tilde{\nabla} dx \\ \langle D_t \mathcal{F}(\dot{y}, p, y, t), (\tilde{z}, \tilde{q}) \rangle &= - \int_{\Omega} \dot{f} \cdot \tilde{v} dx - \int_{\Omega} \dot{h} \cdot \tilde{v} dx. \end{aligned}$$

Proof. I still need to report in $\text{\LaTeX}$ the boring check but it should be true. First we prove the Gateaux differentiability everywhere that ensure also the continuity. Then we prove that the Gateaux differential is continuous. To this aim we use Lemma C.9, the following lemma and the embedding of H^1 in L^6 for $d = 2$ or $d = 3$.

Lemma 2.25. Let $\Omega \subset \mathbb{R}^d$ be open and bounded. Let $\phi : \mathbb{R}^m \rightarrow \mathbb{R}^l$ be Lipschitz continuous and bounded. Let $f_n \rightarrow f$ in $L^p(\Omega, \mathbb{R}^m)$ for some $p \in [1, \infty)$. Then $\phi(f_n) \rightarrow \phi(f)$ in $L^q(\Omega, \mathbb{R}^l)$ for all $q \in [p, \infty)$.

Proof. Let $A_n = \{x \in \Omega \mid |f_n(x) - f(x)| < 1\}$. Since convergence in L^p implies convergence in measure then $|\Omega \setminus A_n| \rightarrow 0$. Now let $q \in [p, \infty)$. Then if we call $L > 0$ the Lipschitz constant of ϕ and $C = \sup_{\mathbb{R}^m} |\phi|$ we have

$$\begin{aligned} \|\phi(f_n) - \phi(f)\|_q^q &= \int_{A_n} |\phi(f_n) - \phi(f)|^q dx + \int_{\Omega \setminus A_n} |\phi(f_n) - \phi(f)|^q dx \leq \\ &\leq \int_{A_n} L |f_n - f|^q dx + 2C |\Omega \setminus A_n|. \end{aligned}$$

But on A_n we have $|f_n - f|^q \leq |f_n - f|^p$ so that

$$\|\phi(f_n) - \phi(f)\|_q^q \leq \int_{A_n} L |f_n - f|^p dx + 2C |\Omega \setminus A_n| \leq L \|f_n - f\|_p^p + 2C |\Omega \setminus A_n|.$$

This proves that $\phi(f_n) \rightarrow \phi(f)$ in $L^q(\Omega, \mathbb{R}^l)$. □

□

We notice that $\mathcal{F}$ does not depend on λ . We call $\xi = (\dot{u}, \dot{U}, \lambda, p)$, $\zeta = (v, V, \mu, q) = (z, q)$ and $\tilde{\zeta} = (\tilde{v}, \tilde{V}, \tilde{\mu}, \tilde{q}) = (\tilde{z}, \tilde{q})$ all belonging to $\mathcal{Z} \times \mathcal{Q}$.

Proposition 2.26. For any $\dot{y} \in \mathcal{Z}$, $p \in \mathcal{Q}$, $y \in \mathcal{Z}$ and $t \in [0, T]$ we have $D_\xi \mathcal{F}(\dot{y}, p, y, t) \in \text{Iso}(\mathcal{Z} \times \mathcal{Q}, \mathcal{Z}' \times \mathcal{Q}')$.

Proof. We define the bilinear form $a: \mathcal{Z} \times \mathcal{Z} \rightarrow \mathbb{R}$ by

$$\begin{aligned} a(z, \tilde{z}) = \int_{\Omega} \mathfrak{D} E v : E \tilde{v} dx - \int_{\Omega} \mathfrak{e} \frac{\mu}{d} I : \left(E \tilde{v} - \tilde{V} - \frac{\tilde{\mu}}{d} I \right) dx + \int_{\Omega} \mathfrak{e} \text{grad}^2 v : \text{grad}^2 \tilde{v} dx + \\ + \int_{\Omega} \mathfrak{F} \nabla \circ \tilde{\nabla} dx + \int_{\Omega} Y(p) \partial^2 G(\dot{U}) V : \tilde{V} dx. \end{aligned}$$

We define also the bilinear forms $b: \mathcal{Q} \times \mathcal{Z} \rightarrow \mathbb{R}$ and $c: \mathcal{Q} \times \mathcal{Z} \rightarrow \mathbb{R}$ by

$$b(p, \tilde{z}) = \int_{\Omega} (Y'(p) \partial G(\dot{U}) : \tilde{V} - \tilde{\mu}) q dx$$

and

$$c(\tilde{q}, z) = - \int_{\Omega} \mu \tilde{q} dx = 0.$$

To prove that $D_\xi \mathcal{F}(\dot{y}, p, y, t) \in \text{Iso}(\mathcal{Z} \times \mathcal{Q}, \mathcal{Z}' \times \mathcal{Q}')$ we have to prove that the mixed elliptic problem

$$\begin{aligned} a(z, \tilde{z}) + b(p, \tilde{z}) &= \langle l, \tilde{z} \rangle \quad \text{for all } \tilde{z} \in \mathcal{Z} \\ c(\tilde{q}, z) &= \langle m, \tilde{q} \rangle \quad \text{for all } \tilde{q} \in \mathcal{Q} \end{aligned}$$

is well posed for all $l \in \mathcal{Z}'$ and $m \in \mathcal{Q}'$. To this aim it is sufficient to check the hypotheses of Theorem C.7. We first check the hypotheses on b . We have

$$\tilde{\mathcal{N}} = \{ \tilde{z} \in \mathcal{Z} \mid b(p, \tilde{z}) = 0 \text{ for all } p \in \mathcal{Q} \} = \{ \tilde{z} \in \mathcal{Z} \mid \tilde{\mu} = Y'(p) \partial G(\dot{U}) : \tilde{V} \}.$$

Notice that $|Y'(p) \partial G(\dot{U}) : \tilde{V}| \leq \text{Lip}(Y) \|\partial G\|_\infty |\tilde{V}| \in L^2$. Now

$$\sup_{q \in \mathcal{Q} \setminus \{0\}} \frac{\int_{\Omega} (Y'(p) \partial G(\dot{U}) : \tilde{V} - \tilde{\mu}) q dx}{\|q\|_Q} = \|Y'(p) \partial G(\dot{U}) : \tilde{V} - \tilde{\mu}\|_2.$$

On the other hand since $\tilde{z}_0 = (\tilde{v}_0, \tilde{V}_0, \tilde{\mu}_0) \in \tilde{\mathcal{N}}$ if and only if $\tilde{\mu}_0 = Y'(p) \partial G(\dot{U}) : \tilde{V}_0$ we have

$$\begin{aligned} \|\tilde{z}\|_{\mathcal{Z}/\tilde{\mathcal{N}}} &= \inf_{\tilde{z}_0 \in \tilde{\mathcal{N}}} [\|\tilde{v} + \tilde{v}_0\|_{1,2} + \|\tilde{V} + \tilde{V}_0\|_{1,2} + \|\tilde{\mu} + Y'(p) \partial G(\dot{U}) : \tilde{V}_0\|_2] = \\ &= \inf_{\tilde{V}_0 \in H^1} [\|\tilde{V} + \tilde{V}_0\|_{1,2} + \|\tilde{\mu} + Y'(p) \partial G(\dot{U}) : \tilde{V}_0\|_2] \leq \\ &\leq \|\tilde{\mu} - Y'(p) \partial G(\dot{U}) : \tilde{V}\|_2 \end{aligned}$$

where the last inequality holds since we can choose $\tilde{V}_0 = -\tilde{V}$. Then we have

$$\sup_{q \in \mathcal{Q} \setminus \{0\}} \frac{\int_{\Omega} (Y'(p) \partial G(\dot{U}) : \tilde{V} - \tilde{\mu}) q dx}{\|q\|_Q} \geq \|\tilde{z}\|_{\mathcal{Z}/\tilde{\mathcal{N}}}.$$

Moreover if $q \in \mathcal{Q}$ is such that

$$b(p, \tilde{z}) = \int_{\Omega} (Y'(p) \partial G(\dot{U}) : \tilde{V} - \tilde{\mu}) q dx = 0 \quad \text{for all } \tilde{z} \in \mathcal{Z}$$

then choosing $\tilde{V} = 0$ and $\tilde{\mu} \in L^2(\Omega)$ arbitrary we have $q = 0$. This completes the check of the hypotheses on b. The check of the hypotheses on c is similar. In particular we have

$$\mathcal{N} = \{z \in \mathcal{Z} \mid c(\tilde{q}, z) = 0 \text{ for all } \tilde{q} \in \mathcal{Q}\} = \{z \in \mathcal{Z} \mid \mu = 0\}.$$

Now we check the hypotheses on a. For any $z = (v, V, 0) \in \mathcal{N}$ we consider $\tilde{z} = (v, V, Y'(p)\partial G(\tilde{U}) : V) \in \tilde{\mathcal{N}}$ and conversely. We notice that with that choice of z and $\tilde{z}$ there exist $K > 0$ and $C > 0$ such that $\|\tilde{z}\|_{\mathcal{Z}} \leq K\|z\|_{\mathcal{Z}}$ and

$$\alpha(z, \tilde{z}) \geq \delta\|Ev\|_2^2 + \epsilon\|\text{grad}^2 v\|_2^2 + \phi\|\mathbb{V}\|_2^2 \geq C\|z\|_{\mathcal{Z}}^2 \geq \frac{C}{K^2}\|\tilde{z}\|_{\mathcal{Z}}^2.$$

This immediately implies that for all $z \in \mathcal{N}$ we have

$$\sup_{\tilde{z} \in \tilde{\mathcal{Z}}} \frac{\alpha(z, \tilde{z})}{\|\tilde{z}\|_{\mathcal{Z}}} \geq \frac{C}{K}\|z\|_{\mathcal{Z}}$$

and if $\tilde{z} \in \tilde{\mathcal{N}}$ satisfy $\alpha(\cdot, \tilde{z}) = 0$ then $\tilde{z} = 0$. Then also the hypotheses on a are verified. This completes the proof. $\square$

By Proposition 2.26 in a neighborhood of every $(\dot{y}_0, p_0, y_0, t_0)$ with $\lambda_0 = 0$ we can explicit $\xi = (\dot{u}, \dot{U}, \lambda, p)$ from the equation

$$\mathcal{F}(\dot{y}, p, y, t) = 0.$$

We get

$$(\dot{u}, \dot{U}) = \mathcal{G}(u, U, t) \quad \text{and} \quad (\lambda, p) = \mathcal{H}(u, U, t)$$

where $\mathcal{G}$ and $\mathcal{H}$ are maps of class C^1 . Then we can apply Theorem C.12 to get locally in time a unique solution (u, U) of class C^2 to the Cauchy problem

$$\begin{cases} (\dot{u}, \dot{U}) = \mathcal{G}(u, U, t) \\ u(0) = u_0 \\ U(0) = U_0. \end{cases}$$

Then we set $(\lambda, p) = \mathcal{H}(u, U, t)$ which is of class C^1 . It is clear that $\mathcal{F}(\dot{y}, p, y, t) = 0$. In conclusion we have obtained the following theorem.

Theorem 2.27. Let $(\dot{y}_0, p_0, y_0, t_0) \in \mathcal{Z} \times \mathcal{Q} \times \mathcal{Z} \times [0, T]$ with $\lambda_0 = 0$. Then there exists a unique C^1 solution (y, p) to Problem 2.6 defined locally around t_0 .

Considering also Theorem 2.7 we obtain global uniqueness.

Corollary 2.28. Assume the hypotheses of Section 2.2.1 and of Section 2.2.6. Then there exists a unique $y \in C^1([0, T], \mathcal{Z})$ and $p \in C^1([0, T], \mathcal{Q})$ solution to Problem 2.6.

2.2.8 Bifurcation analysis

Remark 2.29. This section is not definitive but contains only the ideas that we have collected in the hope of being able to reach a more complete result regarding the phenomenon of localization of deformation.

Some considerations

Since we have proved in Corollary 2.28 that the solution is unique no bifurcation can occur with nonzero viscosity. As the viscosity approaches to zero we expect that the evolution of the system reveals its extreme sensibility on the problem data when strain localization occurs. To explain better the ideas we present the following example that describes what is the behavior that we expect. Consider a perfectly symmetrical specimen under perfectly symmetrical compressive loads. By uniqueness and symmetry in presence of viscosity the specimen will experience a perfectly symmetrical deformation pattern during the entire evolution (a distributed deformation or at most a symmetrical pattern of localized shear bands). However if the loads, the geometry or the material properties of the specimen are not perfectly symmetric then the system experiences an unsymmetrical and completely different deformation pattern (for example an unsymmetrical pattern of shear bands). When viscosity decreases, the threshold beyond which perturbations from the condition of perfect symmetry trigger a localization with a significantly altered pattern is reduced. In some sense in the case of null viscosity the system is sensible also to vanishing perturbations. This suggests a loss of uniqueness at a stage of the loading program also in case of perfect symmetry.

We also expect that the bifurcation is actually associated with a jump in the time evolution. The reasons are the following.

- (i) When passing to the limit $N \rightarrow \infty$ in the Rothe method in Section 2.2.4 we used the viscosity to get the a priori estimates on $\dot{\mathbf{y}}^N$. We were not able to get them without viscosity, using only the $H^1((0, T))$ regularity of the load, as can be done for Von Mises plasticity [22]. This suggests that without viscosity the solution of our model cannot lie in $H^1((0, T))$ but must lie in $BV((0, T))$.
- (ii) Similar models of materials with pressure dependent yield strength exhibit jumps. It is the case of Cam–Clay model studied in [12] or the model studied analytically in [11] coupling the associative Drucker–Prager model with damage.
- (iii) We expect that when the viscosity is small the evolution becomes very fast where the deformation occurs. In the limit of null viscosity this becomes a jump.

Remark 2.30. My desire is to understand better the phenomenon of localization when the viscosity is small but nonzero trying to understand how the sensitivity of the system to imperfection increases with decreasing viscosity. Anyway, if it is useful in some way, I will try to study also the case of zero viscosity. But I would like to avoid the use of the notion of energetic solution. Indeed the choice made by the system at the jump point in the energetic solution approach is determined by a global minimizing approach that seems nonphysical to us. This view is confirmed in [24]. The approach that I could accept is that of balanced viscosity solutions where is the viscosity that drives the evolution at the jumps.

Study on the invertibility of the stiffness operator

In what follows are reported our efforts to understand better the phenomenon of localization. In particular we have confined ourselves to the time discrete problem (based on the implicit Euler scheme). We computed the “stiffness operator” (the one that would be assembled in the numerical approximation of the model with Newton method). It seems to become singular during plastic flow in absence of viscosity suggesting that the system reaches indeed a bifurcation point (where the jump occurs). This analysis should be intended only as a first trial and not as a final result. We hope that the analysis can be extended also to the time continuous case.

We consider $\mathfrak{D}_\eta$, $\mathfrak{E}_\eta$ and $\mathfrak{F}_\eta$ as dependent continuously a viscosity parameter $\eta > 0$ and vanishing for $\eta = 0$. We call $\mathcal{F}_\eta$ the corresponding functional. We consider the discrete in time problem (Problem 2.8). We focus on a single time step and to simplify the notation we set $y = y_{n-1}^N$, $t = t_{n-1}^N$, $p = p_n^N$ and $\dot{y} = \tau_N^{-1}(y_n^N - y_{n-1}^N)$ and $\tau = \tau_N$. Then $y_n^N = y + \tau\dot{y}$. The incremental problem reduces to find $\dot{y} \in \mathcal{Z}$ and $p \in \mathcal{Q}$ such that

$$\mathcal{F}_{(\tau,\eta)}^{(y,t)}(\dot{y}, p) = \mathcal{F}_\eta(\dot{y}, p, y + \tau\dot{y}, t + \tau) = 0.$$

We compute the Fréchet differential $\mathcal{K}_{(\tau,\eta)}^{(y,t)}(\dot{y}, p)$ of $\mathcal{F}_{(\tau,\eta)}^{(y,t)}$ with respect to $(\dot{y}, p)$. It can be written as

$$\langle \mathcal{K}_{(\tau,\eta)}^{(y,t)}(\dot{y}, p)(z, q), (\tilde{z}, \tilde{q}) \rangle = a_{(\tau,\eta)}^{(\dot{y}, p, y, t)}(z, \tilde{z}) + b_{(\tau,\eta)}^{(\dot{y}, p, y, t)}(q, \tilde{z}) + c_{(\tau,\eta)}^{(\dot{y}, p, y, t)}(\tilde{q}, z).$$

We have introduced the bilinear form

$$\begin{aligned} a_{(\tau,\eta)}^{(\dot{y}, p, y, t)}(z, \tilde{z}) &= \int_{\Omega} \mathfrak{D}_\eta E v : E \tilde{v} dx + \int_{\Omega} \tau \mathfrak{C} \left(E v - V - \frac{\mu}{d} I \right) : \left(E \tilde{v} - \tilde{V} - \frac{\tilde{\mu}}{d} I \right) dx + \\ &\quad + \int_{\Omega} (\mathfrak{E}_\eta + \tau \mathfrak{A}) \text{grad}^2 v : \text{grad}^2 \tilde{v} dx + \int_{\Omega} (\mathfrak{F}_\eta + \tau \mathfrak{B}) \nabla \circ \tilde{\nabla} dx + \\ &\quad + \int_{\Omega} Y(p) \partial^2 G(\dot{U}) V : \tilde{V} dx \\ b_{(\tau,\eta)}^{(\dot{y}, p, y, t)}(q, \tilde{z}) &= \int_{\Omega} q(Y'(p) \partial G(\dot{U}) : \tilde{V} - \tilde{\mu}) dx \\ c_{(\tau,\eta)}^{(\dot{y}, p, y, t)}(q, \tilde{z}) &= - \int_{\Omega} \tau \mu \tilde{q} dx. \end{aligned}$$

We want to study if $\mathcal{K}_{(\tau,\eta)}^{(y,t)}$ is in $\text{Iso}(\mathcal{Z} \times \mathcal{Q}, \mathcal{Z}' \times \mathcal{Q}')$. To this aim we can apply Theorem C.7 to the bilinear forms $a_{(\tau,\eta)}^{(\dot{y}, p, y, t)}$, $b_{(\tau,\eta)}^{(\dot{y}, p, y, t)}$ and $c_{(\tau,\eta)}^{(\dot{y}, p, y, t)}$. Reasoning as in the proof of Proposition 2.26 we have that $b_{(\tau,\eta)}^{(\dot{y}, p, y, t)}$ and $c_{(\tau,\eta)}^{(\dot{y}, p, y, t)}$ satisfies the hypotheses. Moreover the sets $\mathcal{N}$ and $\tilde{\mathcal{N}}$ are defined exactly in the same way. We need to check the hypotheses on $a_{(\tau,\eta)}^{(\dot{y}, p, y, t)}$. We reason again as in the proof of Proposition 2.26. For any $z = (v, V, 0) \in \mathcal{N}$ we consider $\tilde{z} = (v, V, Y'(p) \partial G(\dot{U}) : V) \in \tilde{\mathcal{N}}$ and conversely. We have that with that choice of z and $\tilde{z}$ there exist $K > 0$ such that $\|\tilde{z}\|_{\tilde{\mathcal{Z}}} \leq K \|z\|_{\mathcal{Z}}$. Then

$$\begin{aligned} a_{(\tau,\eta)}^{(\dot{y}, p, y, t)}(z, \tilde{z}) &= \int_{\Omega} \mathfrak{D}_\eta E v : E v dx + \int_{\Omega} \tau \mathfrak{C} (E v - V) : \left(E v - V - \frac{Y'(p) \partial G(\dot{U}) : V}{d} I \right) dx + \\ &\quad + \int_{\Omega} (\mathfrak{E}_\eta + \tau \mathfrak{A}) \text{grad}^2 v : \text{grad}^2 v dx + \int_{\Omega} (\mathfrak{F}_\eta + \tau \mathfrak{B}) \nabla \circ \nabla dx + \\ &\quad + \int_{\Omega} Y(p) \partial^2 G(\dot{U}) V : V dx \\ &\leq \delta_\eta \|E v\|_2^2 + \tau \gamma \|E v - V\|_2^2 - 2\tau \kappa \|E v - V\|_2 \|V\|_2 + \\ &\quad + (\epsilon_\eta + \tau \alpha) \|\text{grad}^2 v\|_2^2 + (\phi_\eta + \tau \beta) \|\nabla\|_2^2 + \sigma(p, \dot{U}) \|V\|_2^2. \end{aligned}$$

Here

$$\kappa = \frac{|\mathfrak{C}| \text{Lip}(Y) \|\partial G\|_\infty}{2\sqrt{d}} \quad \text{and} \quad \sigma(p, \dot{U}) = \text{essinf}_\Omega [Y(p) \text{se}(\partial^2 G(\dot{U}))] \geq 0$$

where se denotes the smallest eigenvalue. Then by Young inequality for all $\theta_1 \in (0, \kappa^{-1}\gamma)$

and $\theta_2 \in (0, \infty)$ we have

$$\begin{aligned} \alpha_{(\tau, \eta)}^{(\dot{y}, p, y, t)}(z, \tilde{z}) &\leq \delta_\eta \|Ev\|_2^2 + \tau(\gamma - \kappa\theta_1) \|Ev - V\|_2^2 - \tau \frac{\kappa}{\theta_1} \|V\|_2^2 + \\ &\quad + (\epsilon_\eta + \tau\alpha) \|\text{grad}^2 v\|_2^2 + (\phi_\eta + \tau\beta) \|\mathbb{V}\|_2^2 + \sigma(p, \dot{U}) \|V\|_2^2. \\ &\leq [\delta_\eta + \tau(\gamma - \kappa\theta_1)(1 - \theta_2)] \|Ev\|_2^2 + (\epsilon_\eta + \tau\alpha) \|\text{grad}^2 v\|_2^2 + \\ &\quad + \left\{ \phi_\eta + \sigma(p, \dot{U}) + \tau \left[(\gamma - \kappa\theta_1) \left(1 - \frac{1}{\theta_2} \right) - \frac{\kappa}{\theta_1} + \beta \right] \right\} \|V\|_2^2 + \\ &\quad + (\phi_\eta + \tau\beta) \|\text{grad} V\|_2^2. \end{aligned}$$

The hypotheses on $\alpha_{(\tau, \eta)}^{(\dot{y}, p, y, t)}$ are satisfied if we can find $\theta = (\theta_1, \theta_2) \in (0, \kappa^{-1}\gamma) \times (0, \infty)$ such that

$$\begin{cases} \delta_\eta + \tau(\gamma - \kappa\theta_1)(1 - \theta_2) > 0 \\ \phi_\eta + \sigma(p, \dot{U}) + \tau \left[(\gamma - \kappa\theta_1) \left(1 - \frac{1}{\theta_2} \right) - \frac{\kappa}{\theta_1} + \beta \right] > 0. \end{cases}$$

It $\eta > 0$ then for each θ fixed there exists $\tau > 0$ sufficiently small such that these conditions are satisfied. This connected to the fact that the solution is unique as already established in Section 2.2.7. When $\eta = 0$ then $\delta_\eta = 0$ and $\phi_\eta = 0$. If $\sigma(p, \dot{U}) > 0$ then we can choose $\theta_2 < 1$ so that the first inequality is satisfied and for $\tau > 0$ sufficiently small also the second inequality is satisfied. If instead $\sigma(p, \dot{U}) = 0$ then the inequalities does not depend on τ and can be solved only if $\beta\gamma > \kappa^2$. Indeed from the first we have $\theta_2 < 1$ and from the second we get

$$\frac{1}{\theta_2} < 1 - \frac{\frac{\kappa}{\theta_1} - \beta}{\gamma - \kappa\theta_1}.$$

Then we can find $\theta_2 \in (0, 1)$ if and only if

$$1 - \frac{\frac{\kappa}{\theta_1} - \beta}{\gamma - \kappa\theta_1} > 1 \quad \text{or equivalently} \quad \beta - \frac{\kappa}{\theta_1} > 0.$$

Then we can find $\theta_1 \in (0, 1)$ if and only if $\beta\gamma > \kappa^2$. This means that hardening must be strong enough compared to the sensitivity of the yield strength on pressure (notice that κ is proportional to $\text{Lip}(Y)$).

Remark 2.31. In the case of Von Mises plasticity the stiffness operator is an isomorphism for any positive hardening. This due to the fact that the additional term that comes from the dependence of the yield strength on the plastic pressure is absent.

Intuition about the localization process

We now assume that $\eta = 0$ and that G is a smooth and convex dissipation potential that satisfy $G(\dot{U}) = |\dot{U}| + c$ for all $\dot{U} \in \text{DevSym}$ with $|\dot{U}| > R$ where $c \in \mathbb{R}$ and $R > 0$ are two constants. This gives a plastic force that is independent on the rate and equal to the yield strength when the rate is sufficiently high. So it is a regularization of the rate independent dissipation potential that preserve the existence of a yield limit. We notice that

$$\partial^2 G(\dot{U}) = \frac{|\dot{U}|^2 I - \dot{U} \otimes \dot{U}}{|\dot{U}|^2} \quad \text{for } |\dot{U}| > R$$

so that $\text{se}(\partial^2 G(\dot{U})) = 0$ there. On the contrary $\text{se}(\partial^2 G(\dot{U}))$ must be very high for $\dot{U}$ near the origin if G is a very close approximation of the rate independent dissipation potential $|\cdot|$. We now consider a loading process. When the load is still small the plastic force is under

the yield limit and $\sigma(p, \dot{U})$ is high. By the analysis we have just performed this ensure that $\mathcal{K}_{(\tau, \eta)}^{(y, t)}(\dot{y}, p)$ is invertible and this suggest that the incremental problem is uniquely solved. Moreover we expect that the deformation rate it is not localized in some region but distributed all over the body. As the load increases $se(\partial^2 G(\dot{U}))$ decreases (uniformly in all the body). Then $\mathcal{K}_{(\tau, \eta)}^{(y, t)}(\dot{y}, p)$ is invertible only for small time steps τ . Assuming that the hardening is sufficiently small, we can conclude that $\mathcal{K}_{(\tau, \eta)}^{(y, t)}(\dot{y}, p)$ became singular (or is nonsingular only for very small time increments so that the evolution stagnates). This is the signal that we have reached the bifurcation point. Now the system to restore the invertibility of $\mathcal{K}_{(\tau, \eta)}^{(y, t)}(\dot{y}, p)$ and overcome the singularity point makes $\dot{U}$ small everywhere except that in a small shear band. This cause an elastic unloading elsewhere. Then out of the shear band $se(\partial^2 G(\dot{U}))$ becomes big again. This makes $a_{(\tau, \eta)}^{(\dot{y}, p, y, t)}$ coercive because out of the shear band the plastic strain is held back by the coercivity of $\partial^2 G(\dot{U})$. In the shear band the plastic strain is held back by the surrounding regions through the plastic strain gradient.

2.3 Numerical approximation

Chapter 3

Large deformation model

This chapter is not updated and incomplete. If I have time I will try to get an existence theorem also for the large strain model but I give the priority to understand better the localization of deformation in the small strain model.

3.1 Formulation

3.1.1 Kinematics

Let $\Omega \subset \mathbb{R}^3$ be the reference configuration of the body. The deformation of the body is a map $\chi: \Omega \rightarrow \mathbb{R}^d$. We set $\mathbf{v} = \dot{\chi}$ and $\mathbf{F} = \text{Grad } \chi$ and $\mathbf{L} = \dot{\mathbf{F}}\mathbf{F}^{-1}$. We subdivide $\partial\Omega$ into disjoint parts Γ_D and Γ_N . The plastic distortion is a map $\mathbf{F}_p: \Omega \rightarrow \text{Lin}^+$. We set $\mathbf{L}_p = \dot{\mathbf{F}}_p\mathbf{F}_p^{-1}$ and we assume that $\mathbf{L}_p = \mathbf{D}_p \in \text{Sym}$. We set $\mathbf{U}_p = \text{dev } \mathbf{D}_p$ and $\mathbf{M}_p = \text{Grad } \mathbf{U}_p \cdot \mathbf{F}_p^{-1}$. The elastic distortion is $\mathbf{F}_e = \mathbf{F}\mathbf{F}_p^{-1}$ and we set $\mathbf{L}_e = \dot{\mathbf{F}}_e\mathbf{F}_e^{-1}$.

Lemma 3.1. It holds $\mathbf{L} = \mathbf{L}_e + \mathbf{F}_e\mathbf{D}_p\mathbf{F}_e^{-1}$.

Proof. We have $\mathbf{L} = \dot{\mathbf{F}}\mathbf{F}^{-1} = (\dot{\mathbf{F}}_e\mathbf{F}_p + \mathbf{F}_e\dot{\mathbf{F}}_p)\mathbf{F}_p^{-1}\mathbf{F}_e^{-1} = \dot{\mathbf{F}}_e\mathbf{F}_e^{-1} + \mathbf{F}_e\dot{\mathbf{F}}_p\mathbf{F}_p^{-1}\mathbf{F}_e^{-1} = \mathbf{L}_e + \mathbf{F}_e\mathbf{D}_p\mathbf{F}_e^{-1}$. $\square$

We add the kinematical constrain $\text{tr } \mathbf{D}_p = 0$. To impose this constrain we consider a Lagrange multiplier $p: \Omega \rightarrow \mathbb{R}$. The constrain can be written in weak form as

$$-\int_{\Omega} \text{tr } \mathbf{L}_p \tilde{q} dX = 0 \quad \text{for all } \tilde{q}: \Omega \rightarrow \mathbb{R}.$$

3.1.2 Statics

We choose an internal virtual power of the form

$$\tilde{W} = \int_{\Omega} \mathbf{T} : \tilde{\mathbf{L}}_e dX + \int_{\Omega} \left(\mathbf{T}_p : \tilde{\mathbf{U}}_p - p \text{tr } \tilde{\mathbf{L}}_p + \mathbf{K}_p : \tilde{\mathbf{M}}_p \right) dX.$$

Here $\mathbf{T}$ is the macrostress, $\mathbf{T}_p$ is the microstress which can be assumed deviatoric and $\mathbf{K}_p$ is the microhyperstress. We choose an external virtual power of the form

$$\tilde{\mathcal{P}} = \int_{\Omega} \mathbf{f} \cdot \tilde{\mathbf{v}} dX + \int_{\Gamma_N} \mathbf{h} \cdot \tilde{\mathbf{v}} dA + \int_{\Gamma_D} \mathbf{r} \cdot \tilde{\mathbf{v}} dA.$$

Definition 3.2 (Principle of Virtual Powers). Equilibrium holds when $\tilde{W} = \tilde{\mathcal{P}}$ for all virtual rates.

To get the local version of the equilibrium condition we apply Lemma 3.1 and we write the internal virtual power as

$$\tilde{W} = \int_{\Omega} \mathbf{TF}^{-T} : \text{Grad } \tilde{v} dX + \int_{\Omega} \left(-\mathbf{T}_e : \tilde{\mathbf{L}}_p + \mathbf{T}_p : \tilde{\mathbf{U}}_p - p \text{tr } \tilde{\mathbf{L}}_p + \mathbf{K}_p : \tilde{\mathbf{M}}_p \right) dX$$

where $\mathbf{T}_e = \mathbf{F}_e^T \mathbf{TF}_e^{-T}$ is the Mandel stress. Then we apply the Green theorem

$$\begin{aligned} \tilde{W} = & - \int_{\Omega} \text{Div}(\mathbf{TF}^{-T}) : \text{Grad } \tilde{v} dX + \int_{\partial\Omega} (\mathbf{TF}^{-T})\mathbf{N} \cdot \tilde{v} dA + \\ & + \int_{\Omega} \left\{ [\mathbf{T}_p - \text{dev } \mathbf{T}_e - \text{Div}(\mathbf{K}_p \cdot \mathbf{F}_p^{-T})] : \tilde{\mathbf{U}}_p - \left[\frac{1}{d} \text{tr } \mathbf{T}_e + p \right] \text{tr } \tilde{\mathbf{L}}_p \right\} dX + \\ & + \int_{\partial\Omega} (\mathbf{K}_p \cdot \mathbf{F}_p^{-T} \mathbf{N}) : \tilde{\mathbf{U}}_p dA. \end{aligned}$$

Then the local form of the equilibrium condition is

$$\begin{aligned} \text{Div}(\mathbf{TF}^{-T}) + \mathbf{f} &= 0 \quad \text{in } \Omega \\ (\mathbf{TF}^{-T})\mathbf{N} &= \mathbf{r} \quad \text{in } \Gamma_D \\ (\mathbf{TF}^{-T})\mathbf{N} &= \mathbf{h} \quad \text{in } \Gamma_N \\ \mathbf{T}_p - \text{dev } \mathbf{T}_e - \text{Div}(\mathbf{K}_p \cdot \mathbf{F}_p^{-T}) &= 0 \quad \text{in } \Omega \\ \frac{1}{d} \text{tr } \mathbf{T}_e + p &= 0 \quad \text{in } \Omega \\ \mathbf{K}_p \cdot \mathbf{F}_p^{-T} \mathbf{N} &= 0 \quad \text{in } \partial\Omega. \end{aligned}$$

Appendix A

Notation

A.1 Tensors

$\text{Lin} = \mathbb{R}^{d \times d}$	$d \times d$ matrices
Lin^+	$d \times d$ matrices with positive determinant
Sym	$d \times d$ symmetric matrices
Dev	$d \times d$ deviatoric matrices
DevSym	$d \times d$ symmetric and deviatoric matrices

Appendix B

Modeling in continuum mechanics

Appendix C

Tools from functional analysis

In this chapter we collect some tools of linear and nonlinear analysis used in the main body of this thesis.

C.1 Linear functional analysis

C.1.1 Other

Proposition C.1. Let $\Omega \subset \mathbb{R}^d$ be open and bounded. Let $p \in [1, \infty]$. Let $u_n \rightharpoonup u$ in $W^{1,p}(\Omega)$. Then $u_n \rightarrow u$ in $L^p(\Omega)$.

Proof. Since $u_n \rightharpoonup u$ in $W^{1,p}(\Omega)$ then u_n are bounded in $W^{1,p}(\Omega)$. By Rellich–Kondrakov theorem each subsequence of u_n admits a further subsequence that converges strongly in $L^p(\Omega)$. Since we already know the weak convergence in $W^{1,p}(\Omega)$ then the limit is always u . Then necessarily the whole sequence u_n converges strongly to u in $L^2(\Omega)$. $\square$

C.1.2 The Aubin–Lions–Simon lemma

Theorem C.2 (Aubin–Lions–Simon lemma). Let $T > 0$ and $V \subset H \subset E$ be Banach spaces. Assume that the embedding of V in H is compact and the embedding of H in E is compact. Let p and q in $[1, \infty]$. Then the space

$$\{u \in L^p((0, T), V) \mid \dot{u} \in L^q((0, T), E)\}$$

is compactly embedded in

- (i) $L^p((0, T), H)$ if $p < \infty$
- (ii) $C([0, T], H)$ if $p = \infty$ and $q > 1$.

Proof. See [7, Theorem II.5.16]. $\square$

C.1.3 Generalized mixed elliptic problems

We start with the following theorem.

Theorem C.3 (Babuška). Let V and Q be reflexive Banach spaces. Let $b: Q \times V \rightarrow \mathbb{R}$ be bilinear form. Let $Z = \{z \in V \mid b(q, z) = 0 \text{ for all } q \in Q\}$ and define $W = V/Z$. Assume that b is continuous so that there exists $M > 0$ such that

$$b(q, v) \leq M \|q\|_Q \|v\|_V \quad \text{for all } (q, v) \in Q \times V,$$

Assume also that there exists $\beta > 0$ such that

$$\sup_{q \in Q \setminus \{0\}} \frac{b(q, w)}{\|q\|_Q} \geq \beta \|w\|_W$$

for all $w \in W$ and that if $q \in Q$ satisfies

$$b(q, v) = 0 \quad \text{for all } v \in V$$

then $q = 0$. Then for each $f \in Q'$ there exists a unique $w \in W$ such that

$$b(q, w) = \langle f, q \rangle \quad \text{for all } q \in Q$$

and

$$\|w\|_W \leq \frac{1}{\beta} \|f\|_{Q'}.$$

Proof. Step 1. Consider the linear operator $B: W \rightarrow Q'$ defined by

$$\langle Bw, q \rangle = b(w, q).$$

We notice that

$$|\langle Bw, q \rangle| = \inf_{z \in Z} |b(w + z, q)| \leq M \|q\|_Q \inf_{z \in Z} \|w + z\|_V = M \|q\|_Q \|w\|_W$$

so that $B \in \mathcal{L}(W, Q')$.

Step 2. We prove that B is injective. Indeed if $w \in W$ satisfies $Bw = 0$ we have

$$\beta \|w\|_W \leq \sup_{q \in Q \setminus \{0\}} \frac{\langle Bw, q \rangle}{\|q\|_Q} = 0$$

so that $w = 0$.

Step 3. We prove that $\text{range}(B)$ is closed. Indeed let $f_n = Bw_n$ be such that $f_n \rightarrow f$ in Q' . Then f_n is a Cauchy sequence. Now

$$\beta \|w_n - w_m\| \leq \sup_{q \in Q \setminus \{0\}} \frac{\langle B(w_n - w_m), q \rangle}{\|q\|_Q} \leq \|f_n - f_m\|_{Q'}.$$

Then also w_n is a Cauchy sequence in W and $w_n \rightarrow w$ in W for some $w \in W$. Now by continuity of B we have $Bw = f$ so that $f \in \text{range}(B)$.

Step 4. We prove that $\text{range}(B) = Q'$. Indeed let $(B^\top) = \{q \in Q \mid b(q, v) = 0 \text{ for all } v \in V\} = \{0\}$ by hypotheses. On the other hand $(B^\top) = \text{range}(B)^\perp$. Since Q is reflexive $\text{range}(B) = (\text{range}(B)^\perp)^\perp = Q'$.

Step 5. We have proved that B is a bijection from W to Q' . Then for each $f \in Q'$ there exists a unique $w \in W$ such that $Bw = f$. Moreover

$$\beta \|w\|_W \leq \sup_{q \in Q \setminus \{0\}} \frac{\langle Bw, q \rangle}{\|q\|_Q} = \sup_{q \in Q \setminus \{0\}} \frac{\langle f, q \rangle}{\|q\|_Q} \leq \|f\|_{Q'}.$$

This completes the proof. $\square$

Lemma C.4. Assume that the hypotheses of Theorem C.3 holds. Then

$$\sup_{w \in W \setminus \{0\}} \frac{b(q, w)}{\|w\|_W} \geq \beta \|q\|_Q$$

for all $q \in Q$.

Proof. For each $f \in Q'$ let $w_f \in W$ the solution to

$$b(q, w) = \langle f, q \rangle \quad \text{for all } q \in Q.$$

Notice that $\|f\|_{Q'} \geq \beta \|w\|_W$. Now

$$\|q\|_Q = \sup_{f \in Q' \setminus \{0\}} \frac{\langle f, q \rangle}{\|f\|_{Q'}} = \sup_{f \in Q' \setminus \{0\}} \frac{b(q, w_f)}{\|f\|_{Q'}} \leq \sup_{f \in Q' \setminus \{0\}} \frac{b(q, w_f)}{\beta \|w_f\|_W} \leq \sup_{w \in W \setminus \{0\}} \frac{b(q, w)}{\beta \|w\|_W}.$$

This completes the proof. $\square$

Then we get immediately the following proposition.

Corollary C.5. Assume the hypotheses of Theorem C.3. Then for each $g \in W'$ there exists a unique $p \in Q$ such that

$$b(p, w) = \langle g, w \rangle \quad \text{for all } w \in W$$

and

$$\|p\|_Q \leq \frac{1}{\beta} \|g\|_{W'}.$$

Proof. Apply first Lemma C.4 and then Theorem C.3 with the role of W and Q swapped. $\square$

Remark C.6. Assume that $g \in V'$ is such that $\langle g, z \rangle = 0$ for all $z \in Z$. Then we can define g as an element of W' . Indeed for each $w \in W$ we take a representative $v \in V$ still denoted with w and we define $\langle g, w \rangle = \langle g, v \rangle$. The definition does not depend on the choice of the representative. Moreover

$$\|g\|_{W'} = \sup_{w \in W \setminus \{0\}} \frac{\langle g, w \rangle}{\|w\|_W} = \sup_{v \in V \setminus \{0\}} \frac{\langle g, v \rangle}{\|v\|_V} = \|g\|_{V'}.$$

where we use that since V is reflexive for each $w \in W$ there exists a representative $v \in V$ such that $\|w\|_W = \|v\|_V$ and that for any other representative $u \in V$ we have $\|w\|_W \leq \|u\|_V$. In this case the Corollary C.5 can be formulated as follows. There exists a unique $p \in Q$ such that

$$b(p, v) = \langle g, v \rangle \quad \text{for all } v \in V$$

and

$$\|p\|_Q \leq \frac{1}{\beta} \|g\|_{V'}.$$

The following theorem is the main result of this Section. The result is proved in [26] in the Hilbert spaces setting and we have generalized it for Banach spaces.

Theorem C.7. Let $V, \tilde{V}, Q$ and $\tilde{Q}$ be reflexive Banach spaces. Let $b: Q \times \tilde{V} \rightarrow \mathbb{R}$ be a bilinear form. Let $\tilde{Z} = \{\tilde{z} \in \tilde{V} \mid b(q, \tilde{z}) = 0 \text{ for all } q \in Q\}$ and define $\tilde{W} = \tilde{V} / \tilde{Z}$. Assume that there exists $M > 0$ such that

$$b(q, \tilde{v}) \leq M \|q\|_Q \|\tilde{v}\|_{\tilde{V}} \quad \text{for all } q \in Q \text{ and } \tilde{v} \in \tilde{V}.$$

Assume also that there exists $\beta > 0$ such that

$$\sup_{q \in Q \setminus \{0\}} \frac{b(q, \tilde{w})}{\|q\|_Q} \geq \beta \|\tilde{w}\|_{\tilde{W}} \quad \text{for all } \tilde{w} \in \tilde{W}$$

and that if $q \in Q$ satisfies

$$b(q, \tilde{v}) = 0 \quad \text{for all } \tilde{v} \in \tilde{V}$$

then $q = 0$. Let $c: \tilde{Q} \times V \rightarrow \mathbb{R}$ be a bilinear form. Let $Z = \{z \in V \mid b(\tilde{q}, z) = 0 \text{ for all } \tilde{q} \in \tilde{Q}\}$ and define $W = V/Z$. Assume that there exists $N > 0$ such that

$$c(\tilde{q}, v) \leq N \|\tilde{q}\|_{\tilde{Q}} \|v\|_V \quad \text{for all } \tilde{q} \in \tilde{Q} \text{ and } v \in V.$$

Assume also that there exists $\gamma > 0$ such that

$$\sup_{\tilde{q} \in \tilde{Q} \setminus \{0\}} \frac{c(\tilde{q}, w)}{\|\tilde{q}\|_{\tilde{Q}}} \geq \gamma \|w\|_W \quad \text{for all } w \in W$$

and that if $\tilde{q} \in \tilde{Q}$ satisfies

$$c(\tilde{q}, v) = 0 \quad \text{for all } v \in V$$

then $\tilde{q} = 0$. Let $a: V \times \tilde{V} \rightarrow \mathbb{R}$ be a bilinear form. Assume that there exists a constant L such that

$$a(v, \tilde{v}) \leq L \|v\|_V \|\tilde{v}\|_{\tilde{V}} \quad \text{for all } v \in V \text{ and } \tilde{v} \in \tilde{V}.$$

Assume also that there exists $\alpha > 0$ such that

$$\sup_{\tilde{v} \in \tilde{V} \setminus \{0\}} \frac{a(z, \tilde{v})}{\|\tilde{v}\|_{\tilde{V}}} \geq \alpha \|z\|_V \quad \text{for all } z \in Z$$

and that if $\tilde{z} \in \tilde{Z}$ satisfies

$$a(v, \tilde{z}) = 0 \quad \text{for all } v \in V$$

then $\tilde{z} = 0$. Then for all $f \in \tilde{V}'$ and $g \in \tilde{Q}'$ there exists a unique $(u, p) \in V \times Q$ such that

$$\begin{aligned} a(u, \tilde{v}) + b(p, \tilde{v}) &= \langle f, \tilde{v} \rangle \quad \text{for all } \tilde{v} \in \tilde{V} \\ c(\tilde{q}, u) &= \langle g, \tilde{q} \rangle \quad \text{for all } \tilde{q} \in \tilde{Q}. \end{aligned}$$

Moreover

$$\begin{aligned} \|u\|_V &\leq \frac{1}{\gamma} \left(1 + \frac{L}{\alpha}\right) \|g\|_{\tilde{Q}'} + \frac{1}{\alpha} \|f\|_{\tilde{V}'} \\ \|p\|_Q &\leq \frac{1}{\beta} \left(1 + \frac{L}{\alpha}\right) \|f\|_{\tilde{V}'} + \frac{L\beta}{\gamma} \left(1 + \frac{L}{\alpha}\right) \|g\|_{\tilde{Q}'}. \end{aligned}$$

Proof. By Theorem C.3 there exists a unique $w \in W$ such that

$$c(\tilde{q}, w) = \langle g, \tilde{q} \rangle \quad \text{for all } \tilde{q} \in \tilde{Q}$$

and

$$\|w\|_W \leq \frac{1}{\gamma} \|g\|_{\tilde{Q}'}$$

We identify w with one of its representative in V and we can choose it such that $\|w\|_V = \|w\|_W$. We notice that

$$a(w, \tilde{v}) \leq L \|w\|_V \|\tilde{v}\|_{\tilde{V}}.$$

Then $a(w, \cdot) \in \tilde{V}' \subset \tilde{Z}'$. Then $\langle f, \cdot \rangle - a(w, \cdot) \in \tilde{Z}'$. Then by Theorem C.3 the equation

$$a(z, \tilde{z}) = \langle f, \tilde{z} \rangle - a(w, \tilde{z}) \quad \text{for all } \tilde{z} \in \tilde{Z}$$

has a unique solution $z \in Z$ that satisfies

$$\|z\|_V \leq \frac{1}{\alpha} \left(\|f\|_{\tilde{V}'} + \frac{L}{\gamma} \|g\|_{\tilde{Q}'} \right).$$

Now define $u = w + z \in V$ and we notice that

$$a(u, \tilde{z}) = \langle f, \tilde{z} \rangle \quad \text{for all } \tilde{z} \in \tilde{Z} \quad (\text{C.1.1})$$

and

$$\|u\|_V \leq \frac{1}{\gamma} \left(1 + \frac{L}{\alpha} \right) \|g\|_{\tilde{Q}'} + \frac{1}{\alpha} \|f\|_{\tilde{V}'}.$$

Now we notice by (C.1.1) we have

$$\langle f, \tilde{z} \rangle - a(u, \tilde{z}) = \langle f, \tilde{z} \rangle - a(w, \tilde{z}) - a(z, \tilde{z}) = 0 \quad \text{for all } \tilde{z} \in \tilde{Z}.$$

Then by Remark C.6 we have $\langle f, \cdot \rangle - a(u, \cdot) \in \tilde{W}'$ and the equation

$$b(p, \tilde{v}) = \langle f, \tilde{v} \rangle - a(u, \tilde{v}) \quad \text{for all } \tilde{v} \in \tilde{V}$$

has a unique solution $p \in Q$ that satisfies

$$\|p\|_Q \leq \frac{1}{\beta} (\|f\|_{\tilde{V}'} + L\|u\|_V) \leq \frac{1}{\beta} \left(1 + \frac{L}{\alpha} \right) \|f\|_{\tilde{V}'} + \frac{L\beta}{\gamma} \left(1 + \frac{L}{\alpha} \right) \|g\|_{\tilde{Q}'}.$$

This concludes the proof. $\square$

C.2 Nonlinear functional analysis

C.2.1 Other

Lemma C.8. Let X and Y be Banach spaces. Let U an open subset of X and $F: U \rightarrow Y'$. Let $x \in U$. Assume that there exists $A \in \mathcal{L}(X, Y')$ and for all $v \in X$ there exists $\omega: \mathbb{R} \rightarrow [0, \infty)$ with $\omega(\epsilon) \rightarrow 0$ as $\epsilon \rightarrow 0$ such that

$$\left\| \left\langle \frac{F(x + \epsilon v) - F(x)}{\epsilon} - Av, y \right\rangle \right\| \leq \omega(\epsilon) \|y\|_Y$$

for all $y \in Y$ and $\epsilon \in \mathbb{R}$ sufficiently small. Then F is Gateaux differentiable in x with Gateaux differential A .

Proof. Fix $v \in X$. Then by hypotheses there exists $\omega: \mathbb{R} \rightarrow [0, \infty)$ with $\omega(\epsilon) \rightarrow 0$ as $\epsilon \rightarrow 0$ such that

$$\left\| \frac{F(x + \epsilon v) - F(x)}{\epsilon} - Av \right\|_{Y'} = \sup_{y \in Y \setminus \{0\}} \frac{\left\langle \frac{F(x + \epsilon v) - F(x)}{\epsilon} - Av, y \right\rangle}{\|y\|_Y} \leq \omega(\epsilon).$$

Then

$$\left\| \frac{F(x + \epsilon v) - F(x)}{\epsilon} - Av \right\|_{Y'} \rightarrow 0$$

as $\epsilon \rightarrow 0$. This is exactly the thesis. $\square$

Lemma C.9. Let X and Y be two Banach spaces. Let A_n and A in $\mathcal{L}(X, Y')$ such that for all $x \in X$ and $y \in Y$ we have

$$|\langle A_n x - Ax, y \rangle| \leq \omega_n \|x\|_X \|y\|_Y$$

where $\omega_n \rightarrow 0$. Then $A_n \rightarrow A$ in $\mathcal{L}(X, Y')$.

Proof. We have

$$\|A_n - A\|_{\mathcal{L}(X, Y')} = \sup_{x \in X \setminus \{0\}} \frac{\|A_n x - Ax\|_{Y'}}{\|x\|_X} = \sup_{x \in X \setminus \{0\}} \sup_{y \in Y \setminus \{0\}} \frac{|\langle A_n x - Ax, y \rangle|}{\|x\|_X \|y\|_Y} \leq \omega_n \rightarrow 0.$$

This concludes the proof. $\square$

C.2.2 Implicit function theorem

Theorem C.10. Let X, Y and E be Banach spaces. Let U and open set of $X \times Y$ and $(x_0, y_0) \in U$. Let $F \in C^1(U, E)$ such that $F(x_0, y_0) = 0$ and $D_x F(x_0, y_0) \in \text{Iso}(X, E)$. Then we can find $r > 0$ and $s > 0$ with $B(x_0, r) \times B(y_0, s) \subset U$ such that there exist a unique $\chi \in C^1(B(y_0, s), B(x_0, r))$ satisfying

$$F(\chi(y), y) = 0 \quad \text{for all } y \in B(y_0, s).$$

Moreover if F is of class C^r with $r \geq 2$ also χ is of class C^r .

Proof. See [27]. $\square$

C.2.3 Fixed point theorems

Theorem C.11 (Schauder fixed point theorem). Let X be a Banach space and $C \subset X$ closed and convex. Let $F: C \rightarrow C$ continuous and with compact range. Then F admits a fixed point.

Proof. See [27]. $\square$

C.2.4 Local existence theorem for Cauchy problems in Banach spaces

Theorem C.12. Let X be a Banach space and let U be an open subset of $X \times \mathbb{R}$ with $(x_0, 0) \in U$. Let $G \in C^1(U, X)$. Then there exists $T > 0$ such that the Cauchy problem

$$\begin{cases} \dot{x}(t) = G(x(t), t) & \text{for } t \in [0, T] \\ x(0) = x_0 \end{cases}$$

admits a unique solution $x \in C^2([0, T], X)$.

Proof. Since G is of class C^1 there exists a neighborhood $V \subset U$ of $(x_0, 0)$ such that G and $D_x G$ are bounded in V . Let $M = \sup_V \|G\|_X < \infty$ and $L = \sup_V \|D_x G\|_{\mathcal{L}(X, X)}$. We notice that G is Lipschitz continuous with respect to x and uniformly in t on V with Lipschitz constant L . Let $B = B(x_0, r)$ for some $r > 0$ and $T > 0$ such that $B(x_0, r) \times (-T, T) \subset V$. We introduce the normed space $\mathcal{X} = C^0([0, T], B)$ endowed with the norm $\|x\|_{\mathcal{X}} = \sup_{t \in [0, T]} \|x(t)\|_X$. Since $\mathcal{X}$ is a closed subset of $C^0([0, T], B)$ it is a complete metric space. Now introduce the map

$$T: B \rightarrow C^0([0, T], B)$$

defined by

$$T(x)(t) = x_0 + \int_0^t G(x(s), s) ds.$$

We notice that $\|T(x) - x_0\|_X \leq TM$. Up to choose a smaller T we can assume that $TM < r$ so that actually $T: B \rightarrow B$. Moreover for any x_1 and x_2 in X we have

$$\|T(x_2) - T(x_1)\|_X \leq \int_0^T \|G(x_2(s), s) - G(x_1(s), s)\|_X ds \leq TL\|x_2 - x_1\|_X.$$

Up to choose a smaller T we can assume that $TL < 1$. In this case T is a strict contraction. By the Banach–Caccioppoli fixed point theorem there exists a unique $x \in B$ such that $Tx = x$. Since Tx is of class C^2 then also x is of class C^2 . $\square$

C.2.5 Browder–Minty theorem

The results of this section are taken from [31] and [32].

Let X be a Banach space and $T: X \rightarrow X'$ be a nonlinear operator and $f \in X'$. The Browder–Minty theorem allows to study the existence of solutions to the equation $T(x) = f$ when there is not any variational formulation. The key assumption is that T is coercive and monotone, a property that makes T similar in some sense to an elliptic operator.

Definition C.13. Let X be a Banach space and let $T: X \rightarrow X'$. We say that

- (i) T is bounded if maps bounded sets of X into bounded sets of X' ,
- (ii) T is coercive if

$$\frac{\langle T(x), x \rangle}{\|x\|} \rightarrow \infty \quad \text{as } \|x\| \rightarrow \infty$$

and that

- (iii) T is monotone if

$$\langle T(x) - T(y), x - y \rangle \geq 0 \quad \text{for all } x \text{ and } y \text{ in } X.$$

We now prove a lemma that establish the equivalence between the equation $T(x) = f$ and a variational inequality under the assumption that T is monotone.

Lemma C.14. Let $T: X \rightarrow X'$ be continuous and $f \in X'$. Then if $x \in X$ is a solution to the variational inequality

$$\langle T(y) - f, y - x \rangle \geq 0 \quad \text{for all } y \in X$$

it is also a solution to the equation $T(x) = f$. Moreover if T is monotone then any solution to the equation $T(x) = f$ is also a solution to the variational inequality.

Proof. Let $y = x + tv$ with $t > 0$ and $v \in X$. Then if $x \in X$ is a solution to the variational inequality we have

$$\langle T(x + tv) - f, v \rangle \geq 0.$$

Letting $t \rightarrow 0$ we have $\langle T(x) - f, v \rangle \geq 0$. By arbitrariness of v we have $T(x) = f$. Conversely let $x \in X$ be such that $T(x) = f$. By monotonicity

$$\langle T(y) - f, y - x \rangle = \langle T(y) - T(x), y - x \rangle \geq 0.$$

Then x is a solution to the variational inequality. $\square$

Now we study the equation $T(x) = f$ when the Banach space is finite dimensional. This results will be used to get the general result using the Galerkin method.

Proposition C.15. Let $T: \mathbb{R}^n \rightarrow \mathbb{R}^n$ be continuous and coercive. Then for every $f \in \mathbb{R}^n$ the equation $T(x) = f$ has at least one solution.

Proof. Without loss of generality we can assume that $f = 0$. Indeed otherwise define $S(x) = T(x) - f$ and notice that S is still continuous and coercive. Indeed

$$\frac{S(x) \cdot x}{|x|} = \frac{T(x) \cdot x}{|x|} - \frac{f \cdot x}{|x|} \geq \frac{T(x) \cdot x}{|x|} - |f|.$$

Now define $G: \mathbb{R}^n \rightarrow \mathbb{R}^n$ by $G(x) = x - T(x)$. We notice that

$$G(x) \cdot x = |x|^2 - T(x) \cdot x.$$

Since T is coercive there exists an $r > 0$ such that $|x|^2 - G(x) \cdot x = T(x) \cdot x > 0$ for all $x \in \mathbb{R} \setminus B(0, r)$. We define $\Pi: \mathbb{R}^n \rightarrow \mathbb{C}$ with $C = \overline{B}(0, r)$ by

$$\Pi(x) = \begin{cases} x & \text{if } |x| \leq r \\ r \frac{x}{|x|} & \text{if } |x| > r. \end{cases}$$

We set $H(x) = \Pi(G(x))$ and we notice that $H: C \rightarrow C$ and it is continuous. By Brouwer fixed point theorem there exists $x \in C$ such that $H(x) = x$. If $x \in \mathring{C}$ then $x = H(x) = \Pi(G(x)) = G(x)$ so that $T(x) = 0$. If we prove that $x \notin \mathring{C}$ we have concluded. By contradiction assume that $|x| = r$. Then $|G(x)| \geq r$ and

$$|x|^2 = x \cdot x = H(x) \cdot x = \frac{r}{|G(x)|} G(x) \cdot x < \frac{r}{|G(x)|} |x|^2 \leq |x|^2$$

where we used that $|x|^2 > G(x) \cdot x$ for $|x| \geq r$. This is a contradiction. $\square$

We are now in position to prove the Browder–Minty theorem.

Theorem C.16. Let X be a reflexive and separable Banach space. Let $T: X \rightarrow X'$ be bounded, continuous, coercive and monotone. Then for every $f \in X'$ there exists a solution $x \in X$ to the equation $T(x) = f$.

Proof. As anticipated the proof relies on the Galerkin method.

Step 1. Let $\{y_n\} \subset X$ such that the linear combinations of y_n are dense in X . Let

$$X_n = \text{span} \{y_m\}_{m=1}^n.$$

We define $f_n \in X'_n$ by $\langle f_n, x \rangle = \langle f, x \rangle$ for all $x \in X_n$. We also define $T_n: X_n \rightarrow X'_n$ by $\langle T_n(x), y \rangle = \langle T(x), y \rangle$ for all x and y in X_n .

Step 2. We prove that T_n are still bounded, continuous and coercive. For all $x \in X_n$ we have

$$\|T_n(x)\|_{X'_n} = \sup_{y \in X_n \setminus \{0\}} \frac{\langle T_n(x), y \rangle}{\|y\|_{X_n}} \leq \sup_{y \in X \setminus \{0\}} \frac{\langle T(x), y \rangle}{\|y\|_X} = \|T(x)\|_X$$

so the boundedness of T_n follows from the boundedness of T . Moreover let $x_k \rightarrow x$ in X_n . Then

$$\begin{aligned} \|T_n(x_k) - T_n(x)\|_{X'_n} &= \sup_{y \in X_n \setminus \{0\}} \frac{\langle T_n(x_k) - T_n(x), y \rangle}{\|y\|_{X_n}} \leq \sup_{y \in X \setminus \{0\}} \frac{\langle T(x_k) - T(x), y \rangle}{\|y\|_X} = \\ &= \|T(x_k) - T(x)\|_{X'} \rightarrow 0 \end{aligned}$$

and this proves that T_n is continuous. Moreover for $x \in X_n$

$$\frac{\langle T_n(x), x \rangle}{\|x\|_{X_n}} = \frac{\langle T(x), x \rangle}{\|x\|_X}$$

so the coercivity of T_n follows immediately from the coercivity of T . By C.15 we have that for each n there exists $x_n \in X_n$ that solves $T_n(x_n) = f_n$.

Step 3. We have the estimate

$$\frac{\langle T(x_n), x_n \rangle}{\|x_n\|_X} \leq \frac{\langle T_n(x_n), x_n \rangle}{\|x_n\|_X} = \frac{\langle f_n, x_n \rangle}{\|x_n\|_X} = \frac{\langle f, x_n \rangle}{\|x_n\|_X} \leq \|f\|.$$

Since T is coercive then x_n are bounded in X and since T is bounded then $T(x_n)$ are bounded in X' . Then up to a subsequence

$$\begin{aligned} x_n &\rightharpoonup x \quad \text{in } X \\ T(x_n) &\overset{*}{\rightharpoonup} g \quad \text{in } X' \end{aligned}$$

for some $x \in X$ and $g \in X'$.

Step 4. We first prove that $g = f$. To this aim we fix any y_m and we notice that

$$\langle g - f, y_m \rangle = \lim_{n \rightarrow \infty} \langle T(x_n) - f, y_m \rangle = 0.$$

Then by density we have indeed $g = f$. Next we have

$$\langle T(x_n), x_n \rangle = \langle T_n(x_n), x_n \rangle = \langle g_n, x_n \rangle = \langle g, x_n \rangle \rightarrow \langle g, x \rangle.$$

Step 5. Eventually using the monotonicity of T we have

$$0 \geq \langle T(y) - T(x_n), y - x_n \rangle \rightarrow \langle T(y) - g, y - x \rangle = \langle T(y) - f, y - x \rangle.$$

Applying Lemma C.14 we get the thesis. $\square$

Actually to obtain the same existence result the monotonicity assumption can be weakened.

Definition C.17. Let X be a Banach space and let $T: X \rightarrow X'$. We say that T is pseudomonotone if it is bounded and every time

$$x_k \rightharpoonup x \text{ in } X \quad \text{and} \quad \limsup_{k \rightarrow \infty} \langle T(x_k), x_k - x \rangle \leq 0$$

then

$$\liminf_{k \rightarrow \infty} \langle T(x_k), x_k - y \rangle \geq \langle T(x), x - y \rangle$$

for all $y \in X$.

Theorem C.18. Let X be a separable and reflexive Banach space. Let $T: X \rightarrow X'$ be continuous, coercive and pseudomonotone. Then for every $f \in X'$ there exists $x \in X$ such that $T(x) = f$.

Proof. The proof is exactly the same as the one of Theorem C.16 except to Step 5. We know that $x_k \rightharpoonup x$ in X , that $T(x_k) \overset{*}{\rightharpoonup} f$ in X' and $\langle T(x_k), x_k \rangle \rightarrow \langle f, x \rangle$. Then in particular

$$\langle T(x_k), x_k - x \rangle = \langle T(x_k), x_k \rangle - \langle T(x_k), x \rangle \rightarrow \langle f, x \rangle - \langle f, x \rangle = 0.$$

Since T is pseudomonotone then

$$\liminf_{k \rightarrow \infty} \langle T(x_k), x_k - y \rangle \geq \langle T(x), x - y \rangle$$

for all $y \in X$. But

$$\liminf_{k \rightarrow \infty} \langle T(x_k), x_k - y \rangle = \langle f, x - y \rangle.$$

Then

$$\langle f, x - y \rangle \geq \langle T(x), x - y \rangle$$

for all $y \in X$. By arbitrariness of y we have $T(x) = f$. $\square$

The following proposition establish the relationship between monotone and pseudomonotone operators.

Proposition C.19. Let X be a Banach space. Let $T: X \rightarrow X'$ be continuous, bounded and monotone. Then T is pseudomonotone.

Proof. Let $x_k \rightharpoonup x$ with

$$\limsup_{k \rightarrow \infty} \langle T(x_k), x_k - x \rangle \leq 0.$$

Since T is monotone then

$$\langle T(x_k), x_k - x \rangle \geq \langle T(x), x_k - x \rangle \rightarrow 0.$$

This implies also

$$\liminf_{k \rightarrow \infty} \langle T(x_k), x_k - x \rangle \geq 0.$$

Then

$$\lim_{k \rightarrow \infty} \langle T(x_k), x_k - x \rangle = 0.$$

Now let $y \in X$ and set $y_\epsilon = (1 - \epsilon)x + \epsilon y$ for $\epsilon > 0$. By monotonicity

$$0 \leq \langle T(x_k) - T(y_\epsilon), x_k - y_\epsilon \rangle = \langle T(x_k) - T(y_\epsilon), \epsilon(x - y) + x_k - x \rangle.$$

Rearranging the terms we have

$$\epsilon \langle T(x_k), x - y \rangle \geq \langle T(y_\epsilon), x_k - x \rangle - \langle T(x_k), x_k - x \rangle + \epsilon \langle T(y_\epsilon), x - y \rangle.$$

Passing to the limit $k \rightarrow \infty$ with $\epsilon > 0$ fixed

$$\epsilon \liminf_{k \rightarrow \infty} \langle T(x_k), x - y \rangle \geq \epsilon \langle T(y_\epsilon), x - y \rangle.$$

Dividing by ϵ and passing to the limit $\epsilon \rightarrow 0^+$ we obtain using the continuity of T that

$$\liminf_{k \rightarrow \infty} \langle T(x_k), x - y \rangle \geq \langle T(x), x - y \rangle.$$

Eventually

$$\liminf_{k \rightarrow \infty} \langle T(x_k), x_k - y \rangle = \lim_{k \rightarrow \infty} \langle T(x_k), x_k - x \rangle + \liminf_{k \rightarrow \infty} \langle T(x_k), x - y \rangle \geq 0.$$

This proves that T is pseudomonotone. $\square$

The following result is useful.

Proposition C.20. Let X be a Banach space. Let T_1 and T_2 from X into X' be pseudomonotone operators. Then also $T_1 + T_2$ is pseudomonotone.

Proof. Let $x_k \rightharpoonup x$ with

$$\limsup_{k \rightarrow \infty} \langle T_1(x_k) + T_2(x_k), x_k - x \rangle \leq 0.$$

We claim that

$$\limsup_{k \rightarrow \infty} \langle T_i(x_k), x_k - x \rangle \leq 0 \quad (\text{C.2.1})$$

for both $i = 1$ and $i = 2$. By contradiction

$$\limsup_{k \rightarrow \infty} \langle T_2(x_k), x_k - x \rangle = \epsilon > 0.$$

Up to taking a subsequence we can assume that

$$\lim_{k \rightarrow \infty} \langle T_2(x_k), x_k - x \rangle = \epsilon > 0.$$

Then

$$\limsup_{k \rightarrow \infty} \langle T_1(x_k), x_k - x \rangle \leq -\epsilon < 0. \quad (\text{C.2.2})$$

Then since T_1 is pseudomonotone

$$\liminf_{k \rightarrow \infty} \langle T_1(x_k), x_k - x \rangle \geq 0$$

but this contradicts (C.2.2). Then (C.2.1) holds. Since T_i is pseudomonotone then for all $y \in X$ we have

$$\liminf_{k \rightarrow \infty} \langle T_i(x_k), x_k - y \rangle \geq \langle T_i(x), x - y \rangle.$$

It follows that

$$\liminf_{k \rightarrow \infty} \langle T_1(x_k) + T_2(x_k), x_k - y \rangle \geq \langle T_1(x) + T_2(x), x - y \rangle.$$

This means that $T_1 + T_2$ is pseudomonotone. □

Bibliography

- [1] S. Antman. *Nonlinear Problems of Elasticity*. Springer, 2005.
- [2] D. Bigoni. “Bifurcation and Instability of Non-Associative Elastoplastic Solids”. In: *Material Instabilities in Elastic and Plastic Solids* (2000).
- [3] D. Bigoni. “Localizzazione della deformazione e biforcazione in presenza di leggi costitutive elasto-plastiche incrementali: applicazioni a materiali fragili-coesivi”. PhD thesis. Facoltà di Ingegneria - Università di Bologna, 1991.
- [4] D. Bigoni. *Nonlinear Solid Mechanics - Bifurcation Theory and Material Instability*. Cambridge University Press, 2012.
- [5] D. Boffi, F. Brezzi, and M. Fortin. *Mixed Finite Element Methods and Applications*. Springer, 2013.
- [6] J. Bonet and R.D. Wood. *Nonlinear Continuum Mechanics for Finite Element Analysis*. Cambridge University Press, 2008.
- [7] F. Boyer and P. Fabrie. *Mathematical Tools for the Study of the Incompressible Navier-Stokes Equations and Related Models*. Springer, 2010.
- [8] S.C. Brenner and L.R. Scott. *The Mathematical Theory of Finite Element Methods*. Springer, 2008.
- [9] M. Caponi and V. Crismale. “A rate-independent model for geomaterials under compression coupling strain gradient plasticity and damage”. In: *ESAIM Control Optim. Calc. Var.* (2025).
- [10] P.G. Ciarlet. *Mathematical Elasticity: Three-Dimensional Elasticity*. SIAM, 2021.
- [11] V. Crismale. “Energetic solutions for the coupling of associative plasticity with damage in geomaterials”. In: *Nonlinear Analysis* (2019).
- [12] G. Dal Maso, A. DeSimone, and F. Solombrino. “Quasistatic evolution for Cam-Clay plasticity: a weak formulation via viscoplastic regularization and time rescaling”. In: *Calc Var PDEs* (2011).
- [13] I. Ekeland and R. Témam. *Convex Analysis and Variational Problems*. SIAM, 1999.
- [14] G.A. Francfort and A. Mielke. “Existence results for a class of rate-independent material models with nonconvex elastic energies”. In: *Journal für die reine und angewandte Mathematik* (2006).
- [15] G.A. Francfort and M.G. Mora. “Quasistatic evolution in non-associative plasticity revisited”. In: (2017).
- [16] M. Frémond. *Non-Smooth Thermomechanics*. Springer, 2002.
- [17] M.E. Gurtin. *An Introduction to Continuum Mechanics*. Academic Press, Inc., 1981.
- [18] M.E. Gurtin and L. Anand. “A theory of strain-gradient plasticity for isotropic, plastically irrotational materials. Part I: Small deformations”. In: *International Journal of Plasticity* (2005).
- [19] M.E. Gurtin and L. Anand. “A theory of strain-gradient plasticity for isotropic, plastically irrotational materials. Part II: Finite deformations”. In: *International Journal of Plasticity* (2005).

- [20] M.E. Gurtin, E. Fried, and L. Anand. *The Mechanics and Thermodynamics of Continua*. Cambridge University Press, 2010.
- [21] F. Gurtin M.E. Fried. "Tractions, Balances, and Boundary Conditions for Nonsimple Materials with Application to Liquid Flow at Small-Length Scales". In: *Archive for Rational Mechanics and Analysis* (2006).
- [22] W. Han and Reddy B.D. *Plasticity - Mathematical Theory and Numerical Analysis*. Springer, 2013.
- [23] J.J. Marigo and K. Kazymyrenko. "A micromechanical inspired model for the coupled to damage elasto-plastic behavior of geomaterials under compression". In: *Mechanics and Industry* (2019).
- [24] A. Mielke and R. Rossi. "Balanced-Viscosity Solutions to Infinite-Dimensional Multi-Rate Systems". In: *Arch. Rational Mech. Anal.* (2023).
- [25] A. Mielke and T. Roubíček. *Rate-Independent Systems Theory and Application*. Springer, 2015.
- [26] R.A. Nicolaides. "Existence Uniqueness and Approximation for Generalized Saddle Point Problems". In: *SIAM Journal on Numerical Analysis* (1982).
- [27] L. Nirenberg. *Topics in Nonlinear Functional Analysis*. AMS/Courant Institute of Mathematical Sciences, 2001.
- [28] A. Panteghini and L. Bardella. "On the Finite Element implementation of higher-order gradient plasticity, with focus on theories based on plastic distortion incompatibility". In: *Computer Methods in Applied Mechanics and Engineering* (2016).
- [29] D.D. Pollard and S.J. Martel. *Structural Geology - A Quantitative Introduction*. Cambridge University Press, 2020.
- [30] A. Quarteroni and A. Valli. *Numerical Approximation of Partial Differential Equations*. Springer, 1994.
- [31] M. Renardy and R.C. Rogers. *An Introduction to Partial Differential Equations*. Springer, 2004.
- [32] T. Roubíček. *Nonlinear Partial Differential Equations with Applications*. Springer, 2013.
- [33] I. Vardoulakis and J. Sulem. *Bifurcation Analysis in Geomechanics*. Chapman and Hall, 1995.